

Schrödinger Dynamics as the Unique Minimal First-Response Hamiltonian Projection of AASC Quantum Standing

A Kernel-Forced Same-Scope Quantum-Continuation Exhaustion Theorem from Admissibility, Standing, Projective Hilbert Geometry, Closed State Continuation, and Fixed-Carrier Operator Discipline

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Abstract

This manuscript derives closed autonomous scalar nonrelativistic Schrödinger dynamics as the unique minimal first-response same-scope Hamiltonian projection of AASC quantum standing. The proof does not begin by treating a wavefunction, Hilbert vector, Hamiltonian operator, probability amplitude, mass parameter, or spatial Laplacian as primitive. It first fixes the kernel roles required for any determinate same-domain theorem object: admissibility, standing, reference, and irreversibility. Once a closed quantum-continuation target is used as a determinate same-scope object with fixed carrier, state identity, lawful redescription, update rule, probability standing, and reportability, the kernel is already active.

The structural quantum-standing step is imported, rather than independently reproved. Theorem W9.19 of *Complete Joint Constraint-Kernel Realization Closure for the Wheeler-DeWitt System* begins with an exact six-clause representative certificate c^{QP} and constructs conclusions absent from it: the standing-bearing scalar-orbit quotient, its separated transition weight, descended continuation, descended symmetric-monoidal composition, and their arrow/restriction naturality. W10.25 supplies a nonempty clock-neutral WDW realization of both the raw and derived certificates [10, 11]. Representative Hilbert geometry, a faithful $L^2(\mathbb{R}^d)$ carrier, Wigner implementation, phase lift, strong continuity, Stone generation, operator domains, and the spatial Hamiltonian predicate remain separate downstream projection burdens in this manuscript.

The new Schrödinger-side contribution is the Hamiltonian projection and spatial first-response exhaustion. A faithful nonrelativistic projective-Hilbert projection licenses a minimal flat spatial carrier, closed norm/probability continuation, and fixed operator-domain discipline. A same-scope first-response Hamiltonian candidate H must be internal to that licensed projection, linear on the closed representative state space, equipped with a declared self-adjoint realization, local, scalar under lawful spatial redescription, first nontrivial in spatial variation, compatible with closed standing-preserving continuation, and free of hidden anchors, hidden operator load, arbitrary selectors, nonlinear state-dependent terms, undeclared gauge connections, undeclared spin/internal-role operators, relativistic scope changes, nonlocal memory, bath/open-system ledgers, measurement/update selectors, and boundary-condition laundering. These clauses define $\text{SameScopeQuantumSpatial}_1(H)$, abbreviated below as $\text{SSQ}_1(H)$.

The Hamiltonian-response exhaustion theorem proves that every such candidate has the normal form

$$H = a\Delta + M_V.$$

Here M_V is the multiplication operator associated with a declared scalar potential/source ledger. Coefficient fixation after nonrelativistic action and inertial scale anchoring gives $a = -\hbar^2/(2m)$,

yielding

$$i\hbar\partial_t\psi = \left(-\frac{\hbar^2}{2m}\Delta + M_V\right)\psi.$$

Time-dependent potentials, open-system equations, measurement updates, and driven dynamics are not counterexamples; they are declared non-autonomous, open, update, or effective continuation ledgers. Spin, gauge coupling, relativistic equations, nonlinear Schrödinger equations, Lindblad dynamics, collapse/update theories, nonlocal/fractional operators, and quantum-field structures are not denied. They are typed as skin, declared operator/tensor load, source or potential ledger, internal-role enrichment, relativistic projection, effective scope, open-system ledger, measurement/update regime, nonlocal continuation expansion, boundary-domain enrichment, richer scope, or standing failure unless they collapse to the minimal closed Schrödinger form.

Keywords: AASC; Schrödinger equation; quantum foundations; projective Hilbert space; admissibility; standing; same-scope exhaustion; Hamiltonian normal form; self-adjoint generator; unitary continuation; Laplacian; potential; quantum dynamics; measurement update; open systems; modified quantum mechanics.

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Read This First: Proof Spine

The proof class is *kernel-forced same-scope Hamiltonian projection*. Its correctness is determined by the declared target, hypotheses, proof links, and scope report, not by resemblance to a Hilbert-space-first derivation, canonical quantization route, path-integral presentation, or interpretive account of quantum mechanics.

The proof spine is:

determinate nondegenerate closed autonomous quantum-continuation target
 $\Rightarrow K = (\text{Adm}, \text{Stand}, \text{Ref}, \text{Irr}) \Rightarrow A^+$,

$\Rightarrow$ six-clause representative certificate c^{QP}

$\xRightarrow{\text{W9.19}} (P, q, \bar{p}, \bar{C}, \bar{\boxtimes}; c^{\text{QS}})$

$\Rightarrow$ faithful projection of the derived standing quotient,

$\Rightarrow$ transition-probability-preserving projective continuation

$\Rightarrow$ Wigner implementation and projective phase lift

$\Rightarrow$ strongly continuous unitary branch

$\Rightarrow U_t = e^{-itH/\hbar}, \quad D(H) = D(H^*),$

$\Rightarrow \text{SSQ}_1(H) \Rightarrow H = a\Delta + M_V \Rightarrow a = -\frac{\hbar^2}{2m},$

$\Rightarrow i\hbar\partial_t\psi = \left(-\frac{\hbar^2}{2m}\Delta + M_V\right)\psi.$

This is not a derivation of all quantum physics, not a measurement theory, not a non-autonomous driven-dynamics theorem, not a quantum-field construction, not a derivation of $\hbar$, not a derivation of particle masses, and not a denial of declared richer-scope quantum dynamics. It is a fixed-scope normal-form theorem: once AASC quantum standing is faithfully projected into minimal closed autonomous scalar nonrelativistic projective-Hilbert language, and once no extra same-scope operator load is admitted, Schrödinger dynamics is forced.

Given a closed autonomous scalar projective-Hilbert projection on $L^2(\mathbb{R}^d, dx)$, with transition-standing-preserving ray dynamics, an honest unitary time-flow lift, self-adjoint generator, flat spatial envelope, declared scalar potential multiplication ledger, and no hidden gauge, spin, relativistic, open, update, nonlocal, or boundary load, the minimal first-response Hamiltonian form is

$$H = a\Delta + M_V,$$

equivalently, after coefficient fixation by admitted inertial/action scale anchors,

$$H = -\frac{\hbar^2}{2m}\Delta + M_V.$$

The AASC work is to show why these are exactly the fixed-scope target conditions and to classify every deviation.

Roadmap and exact claim

The core thesis is:

Schrödinger dynamics is the unique minimal first-response same-scope
Hamiltonian projection of closed AASC quantum standing.

The result is not the claim that AASC derives every quantum theory from empty context. It is the narrower fixed-domain necessity claim that W9.19 turns an instantiated representative certificate c^{QP} into a standing-bearing quotient and proves the descent/naturality structure used as the domain of the present manuscript's projective-Hilbert projection. This manuscript then discharges its Wigner–Stone and domain burdens, exhausts the admissible minimal Hamiltonian generators, and obtains Schrödinger form.

The paper separates seven levels:

1. **AASC kernel layer:** $K = (\text{Adm}, \text{Stand}, \text{Ref}, \text{Irr})$.
2. **Quantum-standing role layer:** the exact raw certificate c^{QP} and W9.19's construction of $(P, q, \bar{p}, \bar{C}, \bar{\boxtimes}; c^{\text{QS}})$.
3. **Projective-Hilbert projection layer:** Hilbert representative geometry, inner-product standing, and no wavefunction backflow.
4. **Closed-continuation layer:** norm/probability standing, unitary continuation, and self-adjoint generator form.
5. **Hamiltonian spatial-response exhaustion layer:** $\text{SSQ}_1(H) \Rightarrow H = a\Delta + M_V$.
6. **Scale/mass/potential anchoring layer:** $a = -\hbar^2/(2m)$ and V as scalar potential/source ledger.
7. **Modified-dynamics classification layer:** every non-Schrödinger alternative is typed rather than silently admitted or denied.

The proof center is the load-bearing W9.19 structural import plus the present paper's faithful projection and spatial Hamiltonian-response exhaustion theorem. W10.25 is an occupied WDW realization witness, not a carrier source for the spatial target. The remaining quantum-interior ladder begins from the derived certificate c^{QS} and is restated below at theorem-use level. We write SSQ_1 as a compact abbreviation for the formal predicate $\text{SameScopeQuantumSpatial}_1$.

Part I

Standalone AASC Foundation and Proof-Class Lock

1 Proof-class lock and correctness-relevant criticism

The proof class is kernel-forced AASC quantum-continuation normal form. It is evaluated by its named proof links. A canonical quantization derivation, path-integral derivation, Galilean covariance derivation, or Hilbert-space-first reconstruction may be valuable as a comparator, but it is not the correctness standard for this theorem unless it identifies a failure in the fixed AASC proof chain.

Definition 1.1 (Correctness-relevant criticism). *A criticism is correctness-relevant to this manuscript only if it identifies a false premise, invalid inference, hidden import, scope shift, overreported conclusion, or broken named link in the proof chain: target adequacy to the kernel; AMetric no-selector discipline; quantum predictive quotient architecture; projective/ray-state standing; faithful projective-Hilbert projection; closed reversible state continuation; self-adjoint generator extraction; the predicate $\text{SSQ}_1(H)$; Hamiltonian spatial-response exhaustion; scale/mass coefficient anchoring; potential/source ledger typing; measurement/open-system route closure; modified-quantum-dynamics classification; or reportable-scope discipline.*

Remark 1.2 (Proof-class mismatch). *A demand that the proof be rewritten as canonical quantization, path-integral quantization, an empirical reconstruction, an information-theoretic reconstruction, or Hilbert-space-first textbook quantum mechanics changes the proof class unless it identifies a broken AASC theorem link. AASC is not an overlay placed on top of ordinary Schrödinger theory. It is the fixed-domain governance discipline for theorem objects before a Hamiltonian presentation is licensed.*

Proposition 1.3 (Proof-class boundary). *Within the declared proof class, the relevant correctness question is whether the quantum-continuation target, kernel roles, projective-state role, Hilbert projection, closed continuation, first-response Hamiltonian predicate, spatial normal form, and scale anchoring are valid. Framework familiarity or preference for a standard derivation has no bearing on the theorem unless it locates a mathematical or role-typing failure in those links.*

Proof. A proof object is evaluated relative to its target, carrier, hypotheses, inference links, and reportable scope. Framework preference changes none of these. If an objection changes the target to a different proof class, it asks for a different theorem. If it preserves the target and identifies a broken link, it is correctness-relevant. These alternatives exhaust relevance for this manuscript. $\square$

2 Kernel-neutral target adequacy and the AASC kernel

This section makes the kernel use self-contained. The kernel is not assumed as doctrine; it is the role structure forced by determinate same-domain construction. The Schrödinger theorem uses this structure because a closed quantum-continuation candidate is not raw syntax. It is a theorem-bearing object with target, carrier, state identity, lawful redescription, operator-domain status, continuation rule, and reportable scope.

Definition 2.1 (Kernel-neutral target adequacy). *A theorem-bearing act is target-adequate when it satisfies four conditions.*

1. **Target determinacy:** *the act has a repeatable object of evaluation, not a presentation-dependent trace.*
2. **Step/status evaluability:** *its assertions and transitions are evaluable as advancing, blocking, or failing relative to the same target.*
3. **Act-time finality:** *later repair or reclassification is a distinct act, not retrospective standing for the same act.*
4. **Same-regime fidelity:** *lawful redescription preserves target, carrier, role, standing status, and continuation profile.*

Definition 2.2 (AASC kernel). *The AASC kernel is*

$$K = (\text{Adm}, \text{Stand}, \text{Ref}, \text{Irr}),$$

where *Adm* distinguishes licensed from unlicensed use, *Stand* is reusable theorem-bearing act-status, *Ref* fixes the same-domain target, and *Irr* blocks same-act post-hoc repair of a failed act.

Kernel role	Governance work
Reference	Fixes target, carrier, object identity, and lawful redescription.
Standing	Supplies theorem-bearing or counterexample-bearing force and licensed reuse.
Admissibility	Fixes the boundary between licensed use, support, bookkeeping, repair, scope change, selector import, and reportability.
Irreversibility	Blocks same-act repair, deferred certification, retrospective reclassification, and post-hoc standing creation.

Theorem 2.3 (Target adequacy forces the kernel). *If a theorem-bearing act is target-adequate, then the kernel $K = (\text{Adm}, \text{Stand}, \text{Ref}, \text{Irr})$ is already in force for that act.*

Proof. Target determinacy requires a role that preserves a fixed object across use and redescription; that role is reference. Step/status evaluability requires a role by which an assertion can carry licensed force for later use; that role is standing. Act-time finality requires a role preventing later intervention from changing the status of the same act; that role is irreversibility. Same-regime fidelity requires a boundary under which the previous three roles remain jointly enforceable; that boundary is admissibility. Thus the kernel is not optional terminology added after the fact. It is the minimal governance work already performed by any target-adequate theorem-bearing act. $\square$

Corollary 2.4 (Vocabulary independence). *A critic may reject the names *Adm*, *Stand*, *Ref*, *Irr*, but if the critic preserves determinate target identity, evaluable theorem status, act-time distinction between continuation and repair, and same-regime fidelity, the critic preserves the kernel roles under other names. If any of those roles is denied, the same theorem-bearing quantum-continuation target is no longer preserved.*

Lemma 2.5 (Reference from target determinacy). *Target determinacy forces reference. If no role fixes the repeatable object of evaluation, then redescription can change what is being evaluated while preserving the same marks. The result is a trace or presentation change, not the same theorem-bearing act.*

Proof. A theorem-bearing act must retain a determinate target through reuse, comparison, and lawful redescription. Without such fixation, a later use cannot be distinguished from replacement, relabeling, or change of object. The role that preserves target identity under those operations is reference. $\square$

Lemma 2.6 (Standing from step/status evaluability). *Step/status evaluability forces standing. If an assertion can be evaluated as advancing, blocking, or failing relative to a target, then it must have an act-status that determines whether it may be used in licensed continuation.*

Proof. A raw expression may occur without theorem force. To function inside a proof, it must be eligible or ineligible for downstream use relative to the fixed target. That eligibility is standing. If no such role exists, the expression remains inscription, support, or bookkeeping rather than a theorem-bearing step. $\square$

Lemma 2.7 (Irreversibility from act-time finality). *Act-time finality forces irreversibility. If a later correction can convert a failed act into the same admissible act, then failure and success were not determinate statuses of the original act.*

Proof. A corrected later performance may stand as a new act. It cannot make the earlier failed performance the same admissible act without erasing the distinction between continuation and repair. The governance role that blocks this erasure is irreversibility. $\square$

Lemma 2.8 (Admissibility as joint enforceability boundary). *Same-regime fidelity forces admissibility. A fixed theorem domain must preserve target identity, standing status, act-time finality, and continuation profile under lawful redescription. The boundary under which those classifications remain jointly enforceable is admissibility.*

Proof. If lawful redescription can change which acts are licensed, which targets are fixed, or which failures can be repaired, then the regime has no stable theorem object. If those classifications are preserved, there is already a boundary separating licensed from unlicensed use and continuation from repair. That boundary is admissibility. $\square$

Proposition 2.9 (Anti-weakening burden). *A proposed replacement for the kernel is correctness-relevant only if it preserves the same target, theorem-bearing force, act-time finality, lawful redescription, continuation profile, and reportability while omitting one of the kernel roles and without performing the same governance work under another name.*

Proof. If the replacement loses one of the preservation conditions, it changes the target rather than weakening the kernel. If it restores the lost condition by another selector, classifier, repair channel, or status rule, it has reintroduced the omitted governance work. If it changes no theorem-bearing use, it is bookkeeping. These cases exhaust same-domain weakening attempts. $\square$

3 Fixed-domain consequence layer A^+

The Schrödinger proof uses a compact consequence package of the kernel. Write

$$\text{nondegenerate determinate same-domain use} \Rightarrow K \Rightarrow A^+.$$

Definition 3.1 (AASC consequence layer for quantum continuation). *For a fixed theorem domain, A^+ consists of the following consequences.*

1. **AMetric boundary:** *no metric, probability, Hilbert norm, amplitude, phase, state vector, Hamiltonian, basis, scale, or operator domain has primitive admissibility authority.*
2. **No second same-scope classifier:** *a status factor that changes theorem standing is anchor motion, declared operator/tensor load, or forbidden hidden selector work; if it changes no standing, it is skin/bookkeeping.*
3. **No same-act repair:** *later normalization, projection, collapse, branch selection, measurement record, or decoherence report is a new act or scope change, not retrospective standing for the same closed evolution act.*
4. **Standing as reuse-stable admissibility:** *standing is admissibility preserved under licensed same-domain continuation.*

5. **No representative privilege:** global phase, basis, coordinate, and vector representative are skin unless a declared role makes them standing-bearing.
6. **Report preservation:** a report may not claim more theorem force than the act carries.
7. **No carrier/scope transfer without declaration:** adding spin, gauge connection, relativistic structure, field quantization, bath, stochastic update, measurement selector, nonlocality, or boundary condition changes scope unless declared as anchor or operator/tensor load.

Theorem 3.2 (No independent same-scope quantum status discriminator). *In a fixed nondegenerate quantum-continuation theorem domain, no independent same-scope status discriminator can carry theorem force. Any proposed discriminator is one of: governance-equivalent to the admitted interface, bookkeeping/skin, declared anchor motion, declared operator/tensor load, or forbidden hidden selector work.*

Proof. If the discriminator changes standing-bearing use, it performs admissibility-relevant work. In a fixed domain, such work is already governed by Adm, Stand, Ref, and Irr. If the discriminator preserves those roles exactly, it is governance-equivalent. If it changes target, carrier, source, operator domain, boundary, or continuation, it moves anchors or adds operator/tensor load. If it changes no standing-bearing use, it is skin. The remaining possibility is undeclared same-scope authority, which violates the kernel boundary and is forbidden. $\square$

4 Dependency-inversion exclusion for Schrödinger dynamics

The dependency order is not

$$\text{optional AASC framework} \Rightarrow \text{Schrödinger equation.}$$

The dependency order is

$$\begin{aligned} &\text{determinate nondegenerate closed autonomous quantum-continuation target} \Rightarrow K \Rightarrow A^+ \\ &\Rightarrow \text{quantum-continuation admissibility discipline.} \end{aligned}$$

Definition 4.1 (Quantum-continuation theorem act). *A quantum-continuation theorem act is a use of an update law*

$$\psi(t) \mapsto \psi(t + s)$$

or

$$i\hbar\partial_t\psi = H\psi$$

as a standing-bearing same-scope closed evolution rule on a fixed quantum carrier. Such an act fixes state identity, representative/ray relation, norm/probability standing, operator domain, lawful redescription, continuation rule, and reportable scope.

Theorem 4.2 (Quantum-continuation dependency inversion). *Every nondegenerate same-scope closed quantum-continuation theorem act already instantiates the kernel roles. Therefore AASC discipline is not an external condition imposed on Schrödinger dynamics; it is the governance structure required for the quantum-continuation target to be determinate.*

Proof. A closed quantum-continuation theorem act must fix what state object is being evolved, on which carrier, under which redescription law, with which operator domain, under which continuation rule, and with what reportable status. These are precisely target determinacy, step/status evaluability, act-time finality, and same-regime fidelity. By the target-adequacy theorem, the kernel is active. $\square$

5 Cost of kernel denial in the Schrödinger theorem

Kernel denial does not preserve the same Schrödinger theorem with fewer assumptions. It changes or destroys the determinate closed quantum-continuation object.

Kernel role denied	Schrödinger-specific cost
Reference	No fixed state/ray target; no stable meaning for ψ , $[\psi]$, H , V , Δ , norm, domain, or continuation.
Standing	The equation remains syntax, modeling preference, or comparator language, but lacks theorem-bearing closed-continuation force and licensed downstream use.
Admissibility	No boundary separates closed evolution from measurement update, open-system dynamics, nonlinear/effective models, gauge import, spin import, boundary-condition selection, or hidden Hamiltonian selectors.
Irreversibility	Later normalization, projection, collapse, decoherence, measurement record, or domain repair can be laundered into same-act standing.

Corollary 5.1 (No third position). *There is no position from which one may reject the kernel while still using Schrödinger dynamics as a determinate same-scope closed quantum-continuation theorem object. Either the target remains nondegenerate, in which case the kernel is active, or one of the costs in the table is paid and the same theorem object is lost.*

Part II

AASC Quantum Standing Before Wavefunction Dynamics

6 Admissible quantum interior and quotient architecture

Definition 6.1 (Admissible quantum predictive interior). *An admissible quantum predictive interior is a domain supporting standing-bearing predictive state content whose lawful continuation preserves same-domain identity, whose redescrptions preserve quotient identity, whose transition probabilities carry reusable standing, and whose failed uses cannot be retroactively converted into standing without repair or scope change.*

Definition 6.2 (Quantum presentation and quotient). *A quantum presentation consists of*

$$P_Q = (\Psi, \mathcal{O}, \sim, \text{Cont}_Q),$$

where Ψ is a representative state carrier, $\mathcal{O}$ is the standing-bearing observable/transition structure, $\sim$ is lawful redescription, and Cont_Q is admissible quantum continuation. Its physical quantum content is the quotient $\Psi / \sim$ whenever that quotient exists in the required regularity class.

Theorem 6.3 (Quotient necessity for quantum standing). *If a quantum domain supports reusable predictive standing under lawful redescription, then its standing-bearing content factors through a quotient by admissible redescription.*

Proof. If two lawfully redescribed presentations are not identified in predictive standing, the theory privileges a representative or imports a hidden discriminator. Representative privilege violates reference stability, and hidden discrimination violates admissibility because it supplies unwitnessed same-domain status. Therefore lawfully redescribed states must be identified. Conversely, the quotient retains exactly those distinctions that survive admissible witnesses and continuation. $\square$

Definition 6.4 (Quantum admissibility-bearing quotient architecture). *A quantum admissibility-bearing quotient architecture is*

$$A_Q = (\text{Anc}_Q, \text{Ten}_Q, \text{Skin}_Q, \sim_Q, \text{Cont}_Q),$$

where anchors fix domain, boundary, regime, and admissibility envelope; tensors/operators carry standing-bearing content; skin is representational surplus; $\sim_Q$ is lawful identity-preserving equivalence; and Cont_Q is admissible continuation.

Theorem 6.5 (No fourth same-domain quantum role). *Within a fixed admissibility-bearing quantum presentation, every descriptive element is anchor, tensor/operator content, or skin. No fourth same-domain standing-bearing role survives.*

Proof. If varying an element changes the domain, boundary, regime, carrier, or well-typedness conditions, it is anchor. With anchors fixed, if varying it changes invariant predictive content, observable witness, compatibility relation, or continuation output, it is tensor/operator content. If it changes neither, no admissible downstream use distinguishes it; it is skin. A fourth role would carry standing-bearing force while changing neither domain nor invariant content, which is hidden selector work or bookkeeping. Both are excluded. $\square$

7 Projective/ray-state specialization

The projective comparison class is the closed predictive quantum regime in which AASC quantum standing is analyzed before the concrete Hamiltonian is exhausted. It is not Schrödinger dynamics smuggled into the premises. It fixes the minimal regular setting in which quotient states, redescription, transition weights, independent composition, and closed continuation can be evaluated.

Definition 7.1 (Quantum predictive clause declaration). *The closed quantum predictive interface uses the six clause names **QP-Prob**, **QP-Phase**, **QP-Ray**, **QP-Sep**, **QP-Comp**, and **QP-Cont**. Their exact joint statement, including their representative-level proof objects and the typed symmetric-monoidal quantifier for independent composition, is fixed in the shared Definition 8.1. None of those six clauses asserts that the orbit object carries standing or supplies a descended transition weight, continuation, composition, or naturality theorem.*

Lemma 7.2 (Global phase is skin). *If ψ and $e^{i\theta}\psi$ determine identical transition probabilities and identical continuation behavior, then the factor $e^{i\theta}$ is representational skin.*

Proof. The two representatives differ by a change that preserves every standing-bearing predictive use and every closed continuation output modulo the same quotient. Since it changes no admissible downstream content, it is skin. $\square$

Lemma 7.3 (Relative phase is standing-relevant). *Relative phase cannot be collapsed to skin when it affects interference, transition amplitudes, and downstream standing-bearing predictions.*

Proof. A change in relative phase can alter transition weights and interference terms. Such a change affects standing-bearing predictive content. Therefore it is not skin; it belongs to tensor/operator-level predictive structure after the projective quotient is fixed. $\square$

Theorem 7.4 (Projective-state corollary of W9.19). *Given the representative certificate $c_{\mathfrak{S}}^{\text{QP}}$ with its six named proof coordinates, W9.19 constructs the standing-bearing quotient and proves that state identity is projective/ray identity rather than primitive representative identity:*

$$\psi \sim e^{i\theta} \psi, \quad [\psi] \in \mathbb{P}(\mathcal{H}).$$

Proof. Apply Theorem 8.3. The raw phase-invariance and orbit coequalizer certificates make descent possible; W9.19 proves, using kernel Standing and Reference, that the quotient is exactly the standing object and that its descended transition profiles separate. Relative phase remains standing-bearing whenever it changes the descended transition structure. $\square$

Proposition 7.5 (Projective minimality witness). *A vector-level theory with primitive global phase imports hidden selector structure unless that phase carries observable standing. If it carries no standing, it is skin; if it carries standing, it changes the target.*

Proof. Primitive global phase either affects no standing-bearing use or does. If it affects none, it is skin by the no-fourth-role theorem. If it affects standing, it is no longer the projective quantum target and must be declared as additional structure. Treating it as primitive without declaration is hidden selector import. $\square$

8 W9.19 quantum-standing import and W10.25 realization witness

8.1 Certified representative interface and structural standing closure

Definition 8.1 (Certified **QP/QS** boundary). *Let $\mathfrak{S}$ be a nondegenerate determinate closed quantum-predictive theorem object over a domain U . A representative predictive certificate*

$$c_{\mathfrak{S}}^{\text{QP}} = (X, \Gamma \curvearrowright X, p, \mathcal{C}, \boxtimes, \mathbf{1}, C; \zeta_{\text{prob}}, \zeta_{\text{ph}}, \zeta_{\text{ray}}, \zeta_{\text{sep}}, \zeta_{\text{comp}}, \zeta_{\text{cont}}) \quad (\text{QP-cert})$$

contains a nonempty presentation object X , the lawful scalar-redescription action $\Gamma = \mathbb{C}^\times$, a raw transition-weight family $p : X \times_U X \rightarrow [0, 1]$, a declared comparison class $\mathcal{C}$ with symmetric monoidal product, and a raw closed continuation C . Its six proof coordinates have the following exact burdens:

QP-Prob ζ_{prob} *certifies that p is a standing-bearing representative-level comparison family and is stable under the declared base restrictions and lawful arrows;*

QP-Phase ζ_{ph} *certifies that the Γ -action is lawful redescription and that $p(\gamma x, \gamma' y) = p(x, y)$; it does not declare an orbit to be a standing-bearing state;*

QP-Ray ζ_{ray} *supplies the orbit coequalizer $q : X \rightarrow X/\Gamma$ in the declared regularity class and proves its stability under base restriction; it does not assert standing descent;*

QP-Sep ζ_{sep} *is the raw profile-completeness certificate*

$$[p(x, z) = p(y, z) \text{ and } p(z, x) = p(z, y) \ \forall z] \iff q(x) = q(y);$$

QP-Comp ζ_{comp} certifies on representatives that $\boxtimes$ is independently equivariant in both inputs: for every $(\gamma, \gamma') \in \Gamma \times \Gamma$,

$$\boxtimes(\gamma x, \gamma' y) = (\gamma \gamma') \boxtimes(x, y).$$

Equivalently, it is $(\Gamma \times \Gamma)$ -equivariant when the target Γ -action is pulled back through scalar multiplication $\Gamma \times \Gamma \rightarrow \Gamma$. The same certificate proves that transition weights multiply and that the associator, unitors, and symmetry obey their coherence laws; and

QP-Cont ζ_{cont} certifies that C is Γ -equivariant, preserves p , is monoidal on the declared comparison class, and commutes with restriction and lawful arrows.

Write **QP**($\mathfrak{S}; c_{\mathfrak{S}}^{\text{QP}}$) when these raw data and proofs are supplied. In particular, **QP** does not assume that X/Γ carries state standing, does not supply a descended transition weight or continuation, and does not assume quotient-level composition naturality.

A structural quantum-standing certificate

$$c_{\mathfrak{S}}^{\text{QS}} = (P, q, \bar{p}, \bar{C}, \bar{\boxtimes}; \delta_{\text{stand}}, \delta_{\text{sep}}, \delta_{\text{cont}}, \delta_{\text{comp}}, \delta_{\text{nat}}) \quad (\text{QS-cert})$$

proves that $P = X/\Gamma$ is the standing-bearing state object; q is the universal standing descent and has exactly the scalar-redescription orbits as fibres; $\bar{p}$ is well defined and separates standing distinctions; $\bar{C}$ is the unique standing-preserving continuation with $\bar{C}q = qC$; $\bar{\boxtimes}$ is the induced strong symmetric-monoidal independent-composition law; and all four constructions are natural under lawful arrows and base restriction. Write **QS**($\mathfrak{S}; c_{\mathfrak{S}}^{\text{QS}}$) when this derived certificate is supplied.

The boundary contains no flat spatial carrier, carrier dimension, Laplacian, preferred time, Wigner–Stone lift, Hamiltonian-response predicate, potential, or scale anchor. Those remain projection-side burdens.

Definition 8.2 (Downstream projective-Hilbert realization object). After W9.19 has produced $(P_{\mathfrak{S}}, q_{\mathfrak{S}}, \bar{p}_{\mathfrak{S}}, \bar{C}_{\mathfrak{S}}, \bar{\boxtimes}_{\mathfrak{S}}; c_{\mathfrak{S}}^{\text{QS}})$, a downstream realization first constructs a faithful map

$$\Pi_{\text{NR}} : P_{\mathfrak{S}} \longrightarrow \mathbb{P}L^2(\mathbb{R}^d, dx)$$

that preserves standing, $\bar{p}_{\mathfrak{S}}$, $\bar{C}_{\mathfrak{S}}$, and the declared independent-composition law. It then adds, under separate hypotheses, representative Hilbert geometry, Wigner implementation, phase lift, strong continuity, and a self-adjoint generator with declared domain. The projection’s standing-bearing domain and four preservation laws use W9.19 conclusions and are not data of $c_{\mathfrak{S}}^{\text{QP}}$.

Theorem 8.3 (Imported quantum-standing factorization — W9.19). Let $\mathfrak{S}$ be a nondegenerate determinate admissible quantum predictive interior supplied only with the exact representative certificate $c_{\mathfrak{S}}^{\text{QP}}$ of Definition 8.1. Then W9.19 constructs

$$(P_{\mathfrak{S}}, q_{\mathfrak{S}}, \bar{p}_{\mathfrak{S}}, \bar{C}_{\mathfrak{S}}, \bar{\boxtimes}_{\mathfrak{S}}; c_{\mathfrak{S}}^{\text{QS}}),$$

where the quotient’s standing descent, separation, descended continuation, descended strong symmetric-monoidal composition, and arrow/restriction naturality are conclusions. This is Theorem W9.19(1) of Complete Joint Constraint-Kernel Realization Closure for the Wheeler–DeWitt System [10] and is the load-bearing source expansion used here.

Proof. Direct theorem import from W9.19(1). No independent standing-descent, quotient-separation, descended-continuation, descended-composition, or naturality proof is supplied in this manuscript. $\square$

Theorem 8.4 (Occupied clock-neutral realization witness — W10.25). *The WDW ARC supplies a nonempty positive clock-neutral global realization $\mathfrak{Q}_{10}^{\text{gl}}$ with an explicit raw certificate c_{10}^{QP} and the derived certificate $c_{10}^{\text{QS}} = \text{QClose}(c_{10}^{\text{QP}})$, together with functorial descent and exact marked certified recovery [11]. Its physical carrier is the accepted rank-one WDW line. It is a realization witness, not the spatial carrier of this theorem.*

Remark 8.5 (Exact import boundary). *W9.19 derives projective standing, the separated transition structure, descended independent composition, descended continuation, and their naturality from the raw certificate. W10.25 proves that both sides of this boundary are occupied globally and clock-neutrally. Neither theorem supplies $L^2(\mathbb{R}^d)$, a Laplacian, Wigner implementation, a global Stone group, a nonzero Hamiltonian, $\text{SSQ}_1(H)$, V , $\hbar$, or m .*

Table 1: Reciprocal import and non-export ledger.

Coordinate	Disposition in this manuscript
Exact certified QP/QS boundary	The byte-identical shared definition separates six raw ζ -proofs from five derived δ -conclusions.
Standing quotient and descent/naturality	Imported from W9.19; not assumed in c^{QP} .
Occupied clock-neutral WDW model	Imported from W10.25 as a certificate-bearing realization witness only.
Rank-one WDW carrier	Not embedded in or identified with $L^2(\mathbb{R}^d)$.
Hilbert representative geometry and spatial carrier	Constructed as downstream Schrödinger-side projection burdens.
Wigner lift, phase lift, strong continuity, Stone generator, and domains	Discharged only after faithful projective-Hilbert projection.
Δ , $\text{SSQ}_1(H)$, M_V , $\hbar$, m , and boundary data	Entirely Schrödinger-side burdens; not WDW exports.

9 Quantum-interior theorem ladder and provenance

This section begins from W9.19’s derived standing quotient and $c_{\mathfrak{S}}^{\text{QS}}$. The kernel roles are already forced by the determinate nondegenerate theorem act, and W9.19 is the unique source for the standing-descent and quotient-naturality conclusions used here. Hilbert representative geometry, inner-product reconstruction, projected dynamics, and unitary/self-adjoint generation remain downstream results. The new contribution added in this manuscript is the spatial Hamiltonian normal form.

Theorem 9.1 (QI-1: imported structural quantum standing). *The exact representative certificate yields, by Theorem 8.3, the standing-bearing quotient, separated transition weight, descended composition and continuation, and their naturality. Representative vectors have theorem-bearing force only through that derived quotient role.*

Theorem 9.2 (QI-2: Hilbert-or-denial interior theorem). *A nondegenerate quantum predictive regime with transition-probability standing, separability of predictive distinctions, and independent*

composition either licenses representative Hilbert geometry or pays the denial cost: loss of stable predictive state identity, transition structure, composition, or reusable theorem-bearing state standing.

Denial-cost expansion. The conclusion is not a conventional postulate that Hilbert space must be accepted before the theorem begins. The target already requires reusable transition weights between predictive states, lawful redescription of representatives, separation of standing-relevant distinctions, and composition of independently prepared contents. If no representative geometry is licensed, transition weights lose a stable comparison carrier. If no inner-product-compatible representative structure is admitted, pairwise transition standing cannot be transported through composition without adding an undeclared classifier or context-dependent selector. If composition is preserved by a different carrier, that carrier performs the same representative-geometry work under another name. If none is supplied, predictive distinctions cease to be theorem-bearing reusable contents. Thus denial of the Hilbert representative role does not preserve the same quantum-continuation target with fewer assumptions; it pays a loss of transition comparability, composition, or standing-bearing continuation. $\square$

Theorem 9.3 (QI-3: inner-product reconstruction). *Once projective predictive standing and transition-probability comparison are fixed, representative Hilbert geometry is reconstructed up to the equivalence permitted by lawful redescription. The inner product is not primitive AMetric content; it is the representative geometry through which transition standing is computed.*

Theorem-use reconstruction. The standing-bearing quantity is the transition profile between rays, not the coordinates of a chosen vector. To be reusable, that profile must be invariant under lawful redescription and must separate the predictive distinctions the target treats as standing-relevant. A representative inner-product geometry supplies exactly the invariant transition weight

$$\text{TP}([\psi], [\phi]) = \frac{|\langle \psi, \phi \rangle|^2}{\|\psi\|^2 \|\phi\|^2},$$

while quotienting away global phase and normalization skin. Any alternative presentation that preserves the same transition profile, composition behavior, and redescription invariance is governance-equivalent for this target. Any presentation that does not preserve them changes the target or imports a hidden discriminator. Hence the inner-product representative is licensed as downstream standing geometry, not as primitive AMetric content. $\square$

Theorem 9.4 (QI-4: abstract Schrödinger-form continuation). *Closed reversible admissible update preserving projective standing and independent composition while adding no carriers, no hidden generators, and no measurement/open-system ledger is represented, after faithful Hilbert projection, by a strongly continuous unitary group with a self-adjoint generator:*

$$U_t = e^{-itH/\hbar}, \quad i\hbar \partial_t \psi = H\psi.$$

Remark 9.5 (What is new here). *The new theorem in the present manuscript is not the abstract Stone-form result. The new theorem is the spatial Hamiltonian-response exhaustion*

$$\text{SSQ}_1(H) \Rightarrow H = a\Delta + M_V,$$

and its coefficient discipline under admitted nonrelativistic inertial/action scale anchors. This is the quantum analogue of the Einstein manuscript's curvature-response exhaustion step.

Result	Status in this manuscript
Kernel necessity	Restated self-contained; provenance to the kernel source.
Structural quantum standing	W9.19 constructs five quotient-level conclusions from the exact six-clause raw certificate.
Occupied WDW realization	Imported from W10.25 as a clock-neutral rank-one witness, without carrier transfer.
Hilbert representative geometry	Downstream projective-Hilbert realization burden in this manuscript.
Wigner–Stone bridge	Conventional-facing implementation bridge made explicit in Part V.
Spatial Hamiltonian normal form	New central theorem of the paper.
Coefficient $-\hbar^2/(2m)$	Fixed by admitted nonrelativistic inertial/action scale anchors.
Measurement/open-system classification	Route-closure taxonomy, not a measurement theory.
Paired-projection unification status	Programmatic foothold, not a full joint dynamics theorem.

Part III

Hilbert Projection and Schrödinger-Form Exhaustion

10 AMetric nonselection, Hilbert projection, and endpoint discipline

AASC uses AMetricity in its foundational sense: the AMetric boundary is prior to every admitted metric, Hilbert, probability, operator, or scale role. In this paper AMetricity is the no-selector boundary for projective-Hilbert projection, not a decorative absence claim.

Definition 10.1 (AMetric nonselection). *AMetric nonselection is the pre-admissible non-selection condition. At the AMetric boundary no Hilbert space, inner product, norm, probability, amplitude, basis, phase convention, Hamiltonian, time parameter, spatial coordinate, mass scale, or operator domain has standing. It is not a zero Hilbert metric and not a physical vacuum state.*

Definition 10.2 (Post-admission Hilbert metric). *The Hilbert inner product is a post-admission representative structure*

$$\langle \psi, \phi \rangle.$$

It is not primitive AMetric content. It is licensed only after predictive standing and quotient identity require representative standing-weight geometry.

Theorem 10.3 (AMetric-to-Hilbert projection order). *No wavefunction, Hilbert metric, probability amplitude, Hamiltonian, operator domain, or Schrödinger equation has primitive authority at the AMetric boundary. Hilbert geometry becomes admissible only as a role-fixed projection after standing, reference, quotient identity, and continuation are licensed.*

Proof. A Hilbert expression compares representatives and assigns transition or norm standing. At the AMetric boundary no comparison, representative, probability, scale, or selector is admitted. If a Hilbert expression is used before admission, it is a hidden selector. If it is used after admission, it is an admitted anchor, tensor/operator content, or skin. Therefore Hilbert geometry is downstream of admissibility. A Hamiltonian and its domain are still later because they presuppose the projective-Hilbert projection. $\square$

Theorem 10.4 (Scale import and numerical endpoint discipline). *A dimensional numerical value has AASC standing only relative to an admitted scale anchor, calibration ledger, or comparison carrier. Therefore the present theorem derives form standing and role typing; it does not generate absolute dimensional values for $\hbar$, m , particle masses, coupling constants, or potential scales from the AMetric boundary.*

Proof. Dimensional values compare a quantity against a unit, standard, or calibration procedure. At the AMetric boundary no scale or comparison carrier has standing. Once a scale anchor is admitted, dimensional values may stand relative to that anchor. Thus numerical constants are not boundary-generated outputs in this theorem; they are different endpoint types requiring scale import. $\square$

We use the guiding distinction

form standing $\neq$ absolute scale standing.

The theorem derives the standing of the Schrödinger-form projection. Numerical values of $\hbar$, m , and source/potential scales require calibration anchors.

11 ATS and UEAP control of the Schrödinger projection

The proof uses two control grammars. The anchor/tensor/skin decomposition controls the objects that appear in the theorem. UEAP controls the status of the theorem as a reportable claim. They are not additional proof engines for the Hamiltonian-response result; they prevent role confusion and claim over-promotion.

Definition 11.1 (ATS role typing). *In a fixed admissibility-governed domain, every admissibility-bearing descriptive element is typed as one of three roles. An anchor is an admissibility-defining precondition. A tensor/operator is quotient-level invariant or compatibility content carrying standing-bearing force. Skin is representational surplus with no admissibility-bearing authority. No fourth same-domain role is available without scope change, hidden load, or collapse to bookkeeping.*

Proposition 11.2 (ATS ledger for the Schrödinger projection). *The objects used in the Schrödinger projection theorem have the following role ledger.*

Role	Schrödinger-projection content
Anchors	AMetric boundary discipline; admissibility and standing; projective-Hilbert projection envelope; closed nonrelativistic scope; spatial carrier; time as continuation parameter; operator domain; norm/probability closure; mass and $\hbar$ scale anchors.
Operators/tensors	Ray state $[\psi]$; inner product/transition structure; Hamiltonian generator H ; Laplacian Δ ; scalar potential V ; closed continuation U_t ; probability current when used; relative phase.
Skin	Global phase, basis labels, coordinate labels, representative vector choices, normalization conventions after projective standing is fixed, notation, table layout, and comparator derivation language.

Proof. The listed anchors fix the scope in which the question is well typed. The listed tensor/operator elements change standing-bearing state, generator, transition, source, or continuation content when varied. The listed skin elements can be changed without changing standing-bearing content. This is the fixed-domain ATS partition. $\square$

Corollary 11.3 (ATS reading of $\text{SSQ}_1(H)$). *The predicate $\text{SSQ}_1(H)$ is the Hamiltonian-side ATS discipline in formal form. It forbids anchor motion, undeclared operator load, and skin carrying force. Gauge, spin, relativistic, nonlinear, nonlocal, stochastic, open-system, and measurement-update structures are not declared false; they are excluded only from the minimal closed first-response same-scope class unless retyped as declared anchors or operator/tensor load.*

Definition 11.4 (UEAP status of the main claim). *Let c_S be the claim that*

$$i\hbar\partial_t\psi = \left(-\frac{\hbar^2}{2m}\Delta + M_V\right)\psi$$

is the unique minimal first-response same-scope Hamiltonian projection of closed AASC quantum standing. UEAP audits c_S by fixing its target, carrier, admissible alternatives modulo skin, laundering routes, and reportable status.

UEAP coordinate	Value in this manuscript
Target	Minimal closed nonrelativistic first-response Hamiltonian evolution.
Carrier	AASC kernel; quantum quotient/projective standing; Hilbert projection; closed trajectory theory; unitary/self-adjoint generator support; $SSQ_1(H)$; Hamiltonian-response exhaustion; potential/source ledger discipline.
Scope	Closed nonrelativistic projective-Hilbert state evolution on a declared spatial carrier with fixed operator-domain discipline.
Admissible alternatives	Gauge, spin, relativistic, nonlinear, open-system, collapse/update, non-local, quantum-field, boundary-domain, and effective variants, each typed as skin, declared load, richer projection, open ledger, update regime, nonlocal envelope, or standing failure.
Reportable status	Legitimate theorem within declared closed nonrelativistic minimal projection scope.
Non-reportable promotions	All quantum mechanics; the measurement problem; all QFT; relativistic quantum theory; numerical derivation of $\hbar$ or m ; all modified dynamics false; full quantum-gravity unification.

12 From AASC quantum standing to Hamiltonian field-equation form

Textbook quantum mechanics begins after a Hilbert space and Hamiltonian are already in place:

$$i\hbar\partial_t\psi = H\psi.$$

AASC asks what has to be true before that equation has theorem-bearing standing.

The central question is narrow:

Once AASC quantum standing is faithfully projected into closed nonrelativistic projective-Hilbert language, once state/ray/phase/norm/operator roles are fixed by admissibility-preserving boundary trace rather than primitive representative selection, and once no extra same-scope operator load is admitted, what Hamiltonian generator remains possible?

The answer is Schrödinger form. The proof is an AASC exhaustion argument. A same-scope Hamiltonian evolution must use standing-bearing operator content, preserve lawful redescription, preserve closed norm/probability standing, reject primitive phase or scalar import, and block hidden measurement/open-system repair. When those restrictions are enforced, the spatial response side collapses to two primitive slots:

$$\Delta, \quad V.$$

The kinetic coefficient is then fixed by nonrelativistic action and inertial scale anchoring.

12.1 Why the proof must be explicit

The load-bearing new step is the first-response Hamiltonian exhaustion theorem. It is not a preference for simple Hamiltonians. It is a same-scope AASC normal-form theorem with explicit

type restrictions and case exclusions. Higher-derivative, fractional, nonlocal, gauge, spin, relativistic, nonlinear, stochastic, open-system, and measurement-update structures are not declared false. They are typed. They are richer-scope, higher-response, effective, open, update, mixed-projection, or certificate-bearing structures unless they collapse to the minimal closed Schrödinger form.

12.2 Claim-level separation

The paper separates eight levels: AASC kernel; quantum-standing role; projective-Hilbert projection; transition-probability/projective symmetry; Wigner–Stone representative implementation; Hamiltonian spatial-response exhaustion; scale/mass/potential anchoring; and modified-dynamics classification. The new proof contribution is the spatial normal form and its coupling to the already supported closed quantum-interior theorem chain. The Wigner–Stone material is included to make the conventional-facing bridge explicit: norm preservation alone is not treated as sufficient for linear unitary dynamics.

13 Provenance ledger: expanded here, cited for full audit

This manuscript includes the kernel, quotient, projective-standing, projection, trajectory, and Hamiltonian-exhaustion machinery needed to read the Schrödinger theorem as a standalone AASC result. Prior AASC provenance sources remains cited for fuller proofs, historical dependency lineage, and audit expansion.

Provenance layer	Function in this paper
Physical quotient / GR-QM bookkeeping projection	Supplies the quantum predictive regime, Hilbert-or-denial theorem, inner-product reconstruction, linearity/unitarity/Schrödinger-form theorem, and quantum interior rigidity [6].
Realized physical interior	Supplies the standard physics stack bridge and confirms Hilbert/Schrödinger quantum interior as a downstream realized projection, not primitive proof authority [9].
Same-scope operator exhaustion	Supplies operator-totalization, hidden import, and same-scope operator closure discipline [7].
Self-adjoint extension selection	Supplies fixed-base operator-domain, boundary-condition, and Friedrichs/self-adjoint selection discipline [8].
Present paper	Proves $SSQ_1(H)$, Hamiltonian spatial-response normal form, scale/potential role typing, and modified-quantum-dynamics taxonomy.

14 Projection architecture

Definition 14.1 (Quantum admissibility-bearing quotient architecture). *A quantum admissibility-bearing quotient architecture is*

$$A_Q = (\text{Anc}_Q, \text{Ten}_Q, \text{Skin}_Q, \sim_Q, \text{Cont}_Q),$$

where anchors fix domain and admissibility envelope, tensors/operators carry invariant standing-bearing content, skin is representational surplus, $\sim_Q$ is lawful identity-preserving equivalence, and Cont_Q is admissible continuation.

Definition 14.2 (Projection). *A projection of A_Q to a theorem object X is a role-preserving map*

$$\Pi_X : A_Q \rightarrow X$$

whose kernel consists only of skin-level redundancy relative to the fixed comparison class. Projection is not conversion from one formalism to another. It is quotient-realization of role content in a target presentation.

Definition 14.3 (Quantum projection). *A quantum projection is the image of the AASC quantum standing role in a theorem object supporting projective state identity, transition probabilities, closed continuation, and operator action.*

Definition 14.4 (Projective-Hilbert projection). *A projective-Hilbert projection is a projection*

$$\Pi_H : A_Q \rightarrow (\mathcal{H}, \mathbb{P}(\mathcal{H}), \langle \cdot, \cdot \rangle)$$

that represents predictive comparison, transition standing, state quotient identity, and lawful unitary redescription through Hilbert representative geometry.

Definition 14.5 (Hamiltonian projection). *A Hamiltonian projection is a further projection*

$$\Pi_{Ham} : A_Q \rightarrow (\mathcal{H}, D(H), H, U_t)$$

that represents closed standing-preserving continuation through a self-adjoint generator on a declared domain.

Proposition 14.6 (No formalism-to-formalism primacy). *The Schrödinger equation, if derived, is a theorem object of the Hamiltonian projection. It is not the primitive generator of AASC quantum standing.*

Proof. Quantum standing is already fixed by admissibility, standing, quotient identity, transition structure, and continuation. If the Schrödinger equation were used to authorize those requirements, it would retroactively supply boundary standing from inside the projection. That is Hilbert/Hamiltonian backflow. AASC boundary discipline forbids using a downstream presentation to authorize the standing of the architecture that licensed the presentation. $\square$

Part IV

Faithful Projective-Hilbert Projection

15 Hilbert projection licensing

Definition 15.1 (Faithful projective-Hilbert projection). *A projective-Hilbert projection $\Pi_H : A_Q \rightarrow (\mathcal{H}, \mathbb{P}(\mathcal{H}), \langle \cdot, \cdot \rangle)$ is faithful iff:*

1. *Hilbert representatives are downstream from quantum standing;*
2. *physical state identity is ray/projective identity;*
3. *global phase is skin;*
4. *relative phase is tensor/operator-relevant content when it changes transition standing;*

5. every standing-bearing Hilbert distinction is anchor or tensor/operator content;
6. every non-standing Hilbert distinction is skin;
7. no Hilbert fact retro-authorizes admissibility;
8. no hidden basis, domain, Hamiltonian, probability, or measurement selector is introduced;
9. closed continuation preserves norm/probability standing.

Definition 15.2 (Hilbert anchor, operator/tensor, and skin ledger). *In the faithful projective-Hilbert projection:*

- anchors include the Hilbert carrier, projective quotient, inner-product structure, operator domain, closed nonrelativistic scope, spatial carrier, and scale anchors;
- operators/tensors include H , Δ , V , U_t , transition amplitudes modulo global phase, and probability current when used;
- skin includes global phase, basis labels, coordinate representatives, normalization conventions after probability standing is fixed, and notation.

Remark 15.3 (Projection-anchor scope lock). *Nonrelativisticity, closed-system status, spatial dimension, differentiability/domain class, scalar wavefunction carrier, and scalar potential form are projection anchors in this theorem. They are not generated at the AMetric boundary here. A theory that changes these anchors may be legitimate, but it is not an undeclared same-scope alternative to the present minimal first-response result.*

Theorem 15.4 (No primitive Hilbert backflow). *No wavefunction, Hilbert norm, Hamiltonian, or Schrödinger equation can retro-authorize the standing of the quantum predictive interior whose projection it inhabits.*

Proof. A wavefunction or Hamiltonian is formulated after the projective-Hilbert projection is admitted. If it is used to fix the admissibility of the projection itself, then a downstream representative or operator expression is made into a boundary selector. That would either import an anchor not declared in the AASC architecture, privilege a representative, or repair standing post hoc. Each route violates no-backflow discipline. $\square$

Lemma 15.5 (Basis change as lawful redescription). *Unitary basis changes cannot create or destroy standing-bearing quantum content.*

Proof. A basis choice is presentation skin unless it changes the role content of the theorem object. Lawful unitary redescription preserves inner products, transition weights, and the unitary equivalence class of the generator. Therefore it preserves projective quantum standing. $\square$

Lemma 15.6 (Minimal closed representative compatibility). *In the minimal closed representative projection, standing-preserving reversible continuation is implemented by unitary representative dynamics after projective transition standing and continuity are fixed.*

Proof. This lemma records the target compatibility, not a shortcut from norm preservation alone. Closed continuation preserves ray-level transition standing; reversibility gives projective bijectivity; Wigner implementation supplies unitary or antiunitary representatives; strong one-parameter continuity through the identity selects the unitary branch; Stone's theorem supplies the self-adjoint generator. The detailed conventional-facing bridge is stated in Part V. $\square$

16 The minimal nonrelativistic spatial projection envelope

The Hamiltonian normal-form theorem is not stated over an unspecified quantum carrier. Its target is the scalar, closed, nonrelativistic, local first-response envelope. The following definition is the quantum analogue of the Einstein manuscript's four-dimensional Lorentzian metric-projection scope lock.

Definition 16.1 (Minimal nonrelativistic spatial projection envelope). *A minimal nonrelativistic spatial projection envelope is*

$$\mathcal{E}_{NR} = (Q, \delta, dx, \mathcal{H}, D_0, \text{Aut}_Q),$$

where the primary minimal theorem uses $Q = \mathbb{R}^d$, δ_{ij} is the flat Euclidean spatial comparison anchor, dx is the associated measure, $\mathcal{H} = L^2(\mathbb{R}^d, dx)$, D_0 is a declared dense base domain such as $C_c^\infty(\mathbb{R}^d)$ or the standard Sobolev/core domain after closure, and Aut_Q contains lawful Euclidean spatial redescription compatible with the fixed envelope. Declared regular domains $Q \subseteq \mathbb{R}^d$ are handled by a separate domain-extension corollary requiring explicit boundary/domain/self-adjointness certificates. No gauge connection, spin/internal carrier, curved configuration metric, background vector field, boundary selector, topology selector, nonlocal kernel, stochastic record, or bath ledger is present unless declared as anchor or operator/tensor load.

Remark 16.2 (Spatial projection-anchor scope lock). *Flatness, scalar wavefunction status, closed-system status, nonrelativistic time, spatial dimension, measure, domain/core choice, and absence of gauge/spin/nonlocal/open/update ledgers are projection anchors. Changing them may produce legitimate quantum dynamics, but not an undeclared same-scope alternative to the minimal scalar Schrödinger target.*

Projection element	Status in this theorem
$Q = \mathbb{R}^d$	Primary full-space spatial carrier anchor.
δ_{ij}	Flat spatial comparison anchor.
$L^2(Q, dx)$	Representative Hilbert carrier.
$D(H)$	Operator-domain anchor.
Absence of gauge connection	Minimal-scope condition.
Absence of spin/internal space	Minimal-scope condition.
Declared $Q \subseteq \mathbb{R}^d$	Domain-extension scope requiring boundary/self-adjointness certificate.
Absence of boundary selector	Minimal-scope condition for the primary full-space theorem.
V	Declared scalar potential/source ledger.
$\hbar, m$	Action/inertial scale anchors.

Proposition 16.3 (Obvious enriched alternatives are not counterexamples). *Laplace–Beltrami operators on curved configuration manifolds, magnetic Schrödinger operators, Pauli Hamiltonians, Dirac operators, Aharonov–Bohm topological connections, and self-adjoint boundary-condition variants may be admissible in declared enriched scopes. They are not undeclared same-scope alternatives to the minimal scalar nonrelativistic first-response Hamiltonian projection.*

Proof. Each listed alternative changes an anchor or adds operator/tensor content: curved configuration metric, gauge connection, internal spin space, relativistic representation, topology/boundary data, or extension-selection data. The present theorem excludes only undeclared same-scope imports. Declared enriched theories remain legitimate within their own scopes. $\square$

17 Euclidean scalarity and Galilean covariance

The minimal spatial exhaustion uses Euclidean scalarity, not an undifferentiated appeal to Galilean invariance. Euclidean spatial redescription fixes the form of the undeclared second-order principal symbol: absent additional tensor load, the only scalar principal symbol is proportional to the flat metric, hence to Δ . Galilean covariance has a different proof role. Boosts act projectively on representatives and can produce affine momentum and homogeneous phase terms. Those terms are inertial-frame, boost, or energy-baseline ledgers; they do not supply a new same-carrier scalar kinetic normal form.

Proposition 17.1 (Three-way separation of spatial covariance roles). *In the minimal scalar nonrelativistic projection, Euclidean spatial scalarity fixes the Laplacian slot, Galilean inertial covariance classifies boost-induced affine momentum terms, and projective Galilei representation theory supplies conventional-facing support for the mass/inertial scale anchor. These are distinct proof roles.*

Proof. Rotations and translations preserve the flat spatial comparison anchor and constrain undeclared local scalar differential operators. Boosts change inertial frame and representative phase rather than leaving a fixed-frame Hamiltonian literally invariant as the same operator. Projective Galilei representation theory identifies the mass parameter as central inertial data. Confusing these roles would either overstate invariance or hide frame-motion bookkeeping. Separating them preserves the same-scope target. $\square$

18 Boundary compatibility is not hidden quantum transport

Definition 18.1 (Boundary compatibility). *Boundary compatibility is tensor-level compatibility fixed at admission before spatial transport, time-indexed communication, or interior Hamiltonian exchange is admissibility-relevant. It is not an interior transport process.*

Theorem 18.2 (No hidden boundary-to-interior quantum transport). *A boundary compatibility relation cannot be used as a same-scope hidden transport, repair, nonlocal signaling, or update channel inside the admitted closed projective-Hilbert projection. If it is invoked to move standing-bearing content, it becomes declared operator/tensor load, record ledger, boundary certificate, open-system exchange, or inadmissible hidden repair.*

Proof. Transport is an interior continuation relation. Boundary compatibility is fixed prior to interior Hamiltonian transport. Treating compatibility as transmission confuses boundary status with interior dynamics. If compatibility is used to alter or move standing content inside the admitted projection, it has become operator/tensor load or a continuation ledger and must be declared. If it is not declared, it is hidden repair. $\square$

Corollary 18.3 (Entanglement does not license hidden Hamiltonian repair). *Entangled compatibility may constrain joint standing, but it does not add a hidden same-scope term to H unless declared as operator/tensor load, source ledger, or scope enrichment.*

19 Quantum evolution target class

Definition 19.1 (Hamiltonian evolution candidate). *A Hamiltonian evolution candidate is an equation*

$$i\hbar\partial_t\psi = H\psi,$$

where H is a claimed closed same-scope generator on the licensed projective-Hilbert projection.

Definition 19.2 (Same-scope quantum evolution candidate). *A candidate is same-scope iff it preserves projective state identity, Hilbert carrier, operator domain, norm/probability standing, lawful redescription, closed continuation, nonrelativistic spatial projection, and declared potential/source ledger. If it changes spin, gauge, relativistic structure, open-system status, nonlinear state-dependence, boundary conditions, domain, field quantization, or nonlocality, it is richer/effective/open/update scope unless declared.*

Part V

Trajectory Coherence and Boundary-Trace Dynamics

20 AASC coherent quantum trajectory dynamics

Definition 20.1 (Fixed-scope quantum substrate). *A fixed-scope quantum substrate is a tuple*

$$P_Q = (X, A, S, \rightsquigarrow, \sim, \text{Ten}_Q),$$

where X is the state/ray set, A is the admitted closed quantum scope, S is the standing predicate, $\rightsquigarrow$ is a licensed same-scope transition relation, $\sim$ is standing equivalence, and Ten_Q is tensor/operator content.

Definition 20.2 (Licensed same-scope quantum transition). *A relation $x \rightsquigarrow y$ is a licensed same-scope quantum transition iff y is reached from x by a same-scope construction move admitted by the fixed role and license structure.*

Definition 20.3 (Legal and coherent quantum trajectory). *A legal quantum trajectory is a finite nonempty sequence*

$$\tau_Q = (x_0, x_1, \dots, x_n)$$

with $x_i \rightsquigarrow x_{i+1}$ for every $i < n$. *It is coherent iff every state on the trajectory has standing:*

$$\text{Coherent}_Q(\tau_Q) \iff \text{Legal}_Q(\tau_Q) \text{ and } \forall i \leq n, S(x_i).$$

Definition 20.4 (Quantum failure witness). *A failure witness for τ_Q is an index $i \leq n$ such that $\neg S(x_i)$.*

Proposition 20.5 (Coherence characterization). *For a legal trajectory τ_Q ,*

$$\neg \text{Coherent}_Q(\tau_Q) \iff \exists i \leq |\tau_Q| \neg S(\tau_{Q,i}).$$

Proof. This is immediate from the definition of coherence. A coherent legal trajectory has standing at every state; if a state lacks standing, the trajectory is incoherent. Conversely, if no state lacks standing, the legal trajectory is coherent. $\square$

Theorem 20.6 (Composition closure). *If $\text{Coherent}_Q(\tau_1)$ and $\text{Coherent}_Q(\tau_2)$, and τ_1, τ_2 are composable, then $\text{Coherent}_Q(\tau_1 \circ \tau_2)$.*

Proof. Legal composition follows from composability of legal transitions. Standing holds at every state of both trajectories. Since the joining endpoint is shared, every state in the concatenation has standing. $\square$

Theorem 20.7 (Identity-preserving transformation invariance). *Let τ and τ' be legal quantum trajectories that are pointwise identity-preserving equivalent, meaning their corresponding states have equal tensor/operator content. Then*

$$\text{Coherent}_Q(\tau) \iff \text{Coherent}_Q(\tau').$$

Proof. Standing is determined by tensor/operator content on the fixed substrate. If corresponding states have the same tensor/operator content, standing agrees at each stage. Therefore the universal standing condition defining coherence holds for one trajectory exactly when it holds for the other. $\square$

Theorem 20.8 (No deferred quantum coherence). *Let τ be a legal quantum trajectory and let τ' be a prefix of τ . If $\neg \text{Coherent}_Q(\tau')$, then $\neg \text{Coherent}_Q(\tau)$.*

Proof. A failure witness in the prefix remains a state of every extension. Since coherence requires standing at every state of the full trajectory, the extension cannot be coherent. Later normalization, projection, measurement update, or branch selection cannot delete the earlier failure witness as the same closed lineage. $\square$

Corollary 20.9 (No coherence by skin repair). *Changing basis, notation, global phase, or representation cannot repair a failed quantum trajectory.*

21 Boundary-trace role fixation for quantum dynamics

Definition 21.1 (Quantum continuation domain). *A quantum continuation domain is a tuple*

$$D_Q = (E_Q, G_Q, [L_Q]_{\equiv \text{Loc}}, R_Q, \text{Aut}_Q, \text{Adm}_Q),$$

where E_Q is the evaluated fragment, G_Q is the governance structure, $[L_Q]_{\equiv \text{Loc}}$ is a continuation-locus equivalence class, R_Q is a role-anchor family, Aut_Q is the admissible redescription group, and Adm_Q is the admissibility predicate.

Definition 21.2 (Quantum role-anchor equivalence). *Two role anchors R, R' are role-equivalent, written $R \equiv_{\text{Role}} R'$, iff they preserve identical admissibility-bearing role behavior under lawful same-domain continuation, redescription, boundary comparison, and failure handling.*

Definition 21.3 (Boundary trace). *A boundary trace B for a candidate role-anchor family is an admissibility-preserving asymmetry interface between internally non-selective continuation structure and role-anchor fixation. It is bivalent, stable under identity-preserving redescription, and declared as part of the fixed target domain.*

Definition 21.4 (Boundary-trace fixation).

$$\text{Fix}_{D_Q}(B, [L_Q]_{\equiv \text{Loc}}, [R_L]_{\equiv \text{Role}})$$

holds iff the admissible boundary trace B fixes the role-anchor equivalence class $[R_L]_{\equiv \text{Role}}$ for the continuation-locus class $[L_Q]_{\equiv \text{Loc}}$ without selecting a representative R_L .

Theorem 21.5 (AMetric index non-selection). *At an AMetric boundary, no basis, coordinate, phase convention, Hilbert vector, domain label, energy zero, or Hamiltonian representative is selected by default.*

Proof. The AMetric boundary contains no admitted selector, metric, ranking, phase convention, representative, or domain label. Any attempted default either remains inert or introduces admitted structure. If admitted structure is introduced, the boundary is no longer doing the selection; the declared structure is. Therefore no AMetric default selects a role anchor. $\square$

Theorem 21.6 (Boundary-trace fixation is not representative selection). *Boundary-trace fixation of a role-anchor class neither selects a unique representative nor collapses to a selector operation:*

$$\exists![R_L]_{\equiv \text{Role}} \not\Rightarrow \exists!R_L.$$

Proof. The boundary trace is defined on role behavior up to $\equiv_{\text{Role}}$. If the positive boundary orbit is a singleton equivalence class, the role is fixed. Representatives inside that class remain equivalent under lawful redescription. Selecting one would require a further selector not supplied by the trace. $\square$

Proposition 21.7 (State/phase/norm/operator/potential role fixation). *In the present Hamiltonian-projection theorem, the state role, phase role, norm/probability role, operator-domain role, potential/source role, and measurement/update role must be fixed as role-anchor equivalence classes, not as primitive representatives.*

Proof. Each role participates in standing-bearing continuation. The state role fixes projective identity; the phase role separates global skin from relative standing; the norm/probability role fixes closed predictive standing; the operator-domain role fixes where the generator is defined; the potential/source role fixes scalar source content; and the measurement/update role fixes when closed evolution has been exited. If any were selected by representative privilege, the theorem would import a hidden selector. Boundary-trace fixation provides role-anchor class fixation without representative selection. $\square$

Role	Why representative selection fails	Boundary-trace fixation
State	Vector representative cannot carry primitive standing.	Fixes ray/projective state class.
Phase	Global phase has no predictive standing.	Fixes phase modulo global skin.
Norm/probability	Probability cannot be retro-imported as primitive metric.	Fixes closed norm/probability standing.
Operator domain	Domain choice cannot be hidden in H .	Fixes declared self-adjoint domain class.
Potential/source	V cannot be primitive scalar import.	Fixes scalar potential/source ledger.
Measurement/update	Projection/collapse cannot repair closed evolution.	Types measurement as separate update regime.

22 Closed quantum trajectory theorem

Definition 22.1 (Closed quantum trajectory). *A closed quantum trajectory is a legal trajectory in the faithful projective-Hilbert projection whose states include ray identity, transition-probability standing, norm/probability standing, operator-domain status, and continuation relation. Write it as*

$$\tau_{Q,H} = (([\psi_0], D(H)_0), \dots, ([\psi_n], D(H)_n)).$$

Definition 22.2 (Closed norm/probability ledger). *A norm/probability ledger is closed iff every standing-bearing probability exchange relevant to the state evolution is included in the closed continuation rule or declared as open, update, effective, boundary, or record exchange.*

Definition 22.3 (Transition-probability standing). *For nonzero representatives $\psi, \phi \in \mathcal{H}$, transition standing is represented by*

$$\text{TP}([\psi], [\phi]) = \frac{|\langle \psi, \phi \rangle|^2}{\|\psi\|^2 \|\phi\|^2}.$$

It is ray-level content: changing a representative by nonzero scalar phase/normalization does not change the transition standing.

Definition 22.4 (Closed ray-geometry automorphism). *A closed autonomous quantum-continuation act is a ray-geometry automorphism when it transports the internal transition geometry of the closed state space. This means that for rays transported by the same closed continuation,*

$$\text{TP}(\Phi_t([\psi]), \Phi_t([\phi])) = \text{TP}([\psi], [\phi]).$$

This is not the claim that the probability of an evolved state relative to an externally fixed measurement context is constant. Measurement-relative probabilities belong to a context, observable, apparatus, or record ledger. The closed target preserves transition standing under simultaneous same-scope transport of admissible rays.

Theorem 22.5 (Closed quantum ledger coherence). *If a quantum trajectory is coherent and the norm/probability ledger is closed, then*

$$\|\psi(t)\|^2 = \|\psi(0)\|^2$$

for normalized representatives, and transition probabilities are preserved along closed reversible same-scope continuation.

Proof. A coherent trajectory preserves standing at every stage. In the faithful representative projection, norm/probability and transition probability are the ledgers of closed predictive standing. For a closed autonomous ray-geometry automorphism, the relevant invariant is the transition standing between rays transported by the same same-scope continuation, not the probability assigned to a transported state by an externally fixed measurement context. If internal transition standing changed without a declared observable, apparatus, environment, or record ledger, predictive content would enter or leave the closed state geometry without an admissible transition. That is hidden operator load, omitted boundary/open exchange, update import, or standing failure. Since the trajectory is coherent and the ledger is closed, no such witness is present. $\square$

Corollary 22.6 (Open and update caveat). *If norm/probability or transition standing is not preserved in a presented model, then at least one of the following holds: the system is open; a measurement/update regime is active; a sink/source ledger is omitted; non-Hermitian/effective dynamics is declared; scope has changed; or standing failure has occurred. It is not a counterexample to the closed same-scope theorem.*

23 Projective symmetry, Wigner implementation, and closed unitary continuation

The bridge from projective predictive standing to representative Hilbert dynamics must be explicit. Norm preservation alone is not sufficient: nonlinear maps can preserve a norm. The standing-bearing object is the projective transition structure. The representative linear/unitary object is licensed only after the projective symmetry bridge is fixed.

Definition 23.1 (Projective transition-preserving continuation). *A bijection*

$$\Phi_t : \mathbb{P}(\mathcal{H}) \rightarrow \mathbb{P}(\mathcal{H})$$

is a transition-preserving projective continuation iff, for all nonzero ψ, ϕ ,

$$\text{TP}(\Phi_t([\psi]), \Phi_t([\phi])) = \text{TP}([\psi], [\phi]).$$

It is closed when no measurement, bath, stochastic, source/sink, or record ledger is declared.

Theorem 23.2 (Wigner implementation bridge). *Let $\mathcal{H}$ be a complex Hilbert space with $\dim \mathcal{H} \geq 2$, and let*

$$\Phi : \mathbb{P}(\mathcal{H}) \rightarrow \mathbb{P}(\mathcal{H})$$

be a bijection on the full ray space of nonzero vectors modulo nonzero scalar multiplication. For nonzero representatives define

$$\text{TP}([\psi], [\phi]) = \frac{|\langle \psi, \phi \rangle|^2}{\|\psi\|^2 \|\phi\|^2}.$$

If

$$\text{TP}(\Phi[\psi], \Phi[\phi]) = \text{TP}([\psi], [\phi])$$

for all nonzero $\psi, \phi \in \mathcal{H}$, then Φ is implemented on representative Hilbert space by either a unitary or an antiunitary operator, unique up to phase on representatives [15, 14].

Proof. This is the conventional Wigner symmetry theorem under the stated full-ray-bijection and transition-probability hypotheses. The theorem is not being applied to a partial update, a nonlinear norm-preserving map on the unit sphere, or a measurement-relative probability assignment against a fixed external context. The AASC reading is that ray transition standing is the theorem-bearing object, while the unitary or antiunitary map is representative implementation. Zero vectors are excluded because they do not determine rays. Global phase remains skin; transition probabilities, relative phase structure, and composition behavior carry standing. $\square$

Lemma 23.3 (Continuous time continuation selects the unitary branch). *A strongly continuous one-parameter closed time-continuation group satisfying*

$$\Phi_0 = I, \quad \Phi_{t+s} = \Phi_t \Phi_s$$

cannot move continuously from the identity into the antiunitary branch. Therefore the representative implementation of same-scope continuous time evolution is unitary.

Proof. The identity implementation is unitary. The unitary and antiunitary alternatives occupy distinct implementation types: linear complex-unitary versus conjugate-linear norm-preserving. A strongly continuous one-parameter group through the identity cannot switch type without breaking continuity or the group law. Antiunitary transformations may represent declared discrete symmetries such as time reversal; they are not the same-scope continuous time-flow generator. $\square$

Lemma 23.4 (Projective phase-lift and cocycle discipline). *Let Φ_t be a strongly continuous one-parameter projective ray automorphism preserving transition standing. After Wigner implementation and selection of the unitary branch, the representatives may initially satisfy*

$$U_{t+s} = e^{i\alpha(t,s)} U_t U_s.$$

For the closed time-translation group $\mathbb{R}$, a continuous phase choice may be made so that the representative family is an honest unitary group,

$$\tilde{U}_{t+s} = \tilde{U}_t \tilde{U}_s.$$

Any remaining homogeneous phase convention is an energy-baseline ledger and changes no projective standing unless an energy-reference anchor is declared.

Proof. Wigner implementation is unique only up to representative phase. The possible failure of the strict group law is therefore a scalar two-cocycle. For the one-parameter abelian time-translation group $\mathbb{R}$, the continuous scalar multiplier is removable by a continuous phase convention; equivalently, the relevant continuous projective multiplier for this one-parameter time-flow is cohomologically trivial for the present purpose [16, 17]. This scalar ambiguity can therefore be absorbed by rephasing the implementing operators. The remaining freedom is multiplication by a homogeneous phase $e^{-iCt/\hbar}$, which corresponds to shifting the generator by CI . Such a shift is a baseline-energy convention unless an external energy reference gives it standing-bearing force. Thus the projective time action has a phase-fixed unitary representative group. Stone's theorem applies to this rephased group, while the residual CI freedom remains typed as homogeneous energy-baseline ledger rather than new same-scope dynamics. $\square$

Theorem 23.5 (Unitary continuation compatibility). *Closed reversible quantum continuation preserving transition-probability standing is represented by a strongly continuous unitary one-parameter group*

$$U_t^* U_t = I, \quad U_{t+s} = U_t U_s,$$

after a continuous phase choice, with residual homogeneous phase recorded as an energy-baseline ledger.

Proof. Closed trajectory coherence preserves transition standing under simultaneous same-scope ray transport. Reversibility gives projective bijectivity and the projective group law. The Wigner implementation bridge gives unitary or antiunitary representatives; strong continuity through the identity selects the unitary branch. The projective phase-lift lemma then fixes representatives satisfying the honest unitary group law, with any residual homogeneous phase recorded as energy-baseline ledger. Thus closed reversible continuation is represented by a phase-fixed unitary one-parameter group. $\square$

Theorem 23.6 (Self-adjoint generator compatibility). *For strongly continuous closed reversible unitary continuation, there exists a self-adjoint generator H such that*

$$U_t = e^{-itH/\hbar}.$$

Proof. This is Stone's theorem applied to the strongly continuous one-parameter unitary group, with $\hbar$ introduced as the admitted action/phase scale anchor. In units $\hbar = 1$, the QI-4 theorem-use result states $U_t = e^{-itH}$ and $d\psi(t)/dt = -iH\psi(t)$ [6, 12]. $\square$

Corollary 23.7 (Abstract Schrödinger form). *For differentiable vectors in $D(H)$,*

$$i\hbar\partial_t\psi = H\psi.$$

This is the abstract closed-continuation form. The spatial first-response theorem below determines the minimal scalar nonrelativistic normal form of H .

Part VI

Same-Scope First-Response Hamiltonian Candidates

24 The predicate $\text{SSQ}_1(H)$

Theorem 24.1 ($\text{SSQ}_1(H)$ as quantum target translation). *The predicate $\text{SSQ}_1(H)$ is the Hamiltonian-response translation of the fixed AASC target, not a list tailored after the conclusion. Each clause blocks a specific failure of the declared theorem object.*

Proof. Hilbert internality blocks anchor motion outside the licensed projection. Projective compatibility blocks representative vector or global phase privilege. Linearity blocks hidden state-dependent operator load. Self-adjointness blocks closed norm/probability failure. Locality blocks undeclared nonlocal continuation import. Spatial scalarity blocks preferred-direction or frame selectors. First-response order blocks higher-derivative/effective operator load. Domain discipline blocks boundary-condition laundering. The exclusions of gauge, spin, relativistic, nonlinear, open-system, measurement, and nonlocal terms block undeclared richer-scope structure. The potential-ledger clause blocks primitive scalar import. $\square$

Predicate burden	Failure blocked
Hilbert internality	Anchor motion outside $(\mathcal{H}, D(H), \langle \cdot, \cdot \rangle)$.
Projective compatibility	Global phase or representative vector carrying primitive force.
Linearity	Nonlinear state-dependent hidden operator/source load.
Self-adjointness	Closed norm/probability failure.
Locality	Hidden nonlocal continuation import.
Spatial scalarity	Preferred direction, frame, tensor, or basis import.
First-response order	Higher-derivative/effective operator load.
No arbitrary operator functions	Selector import.
No extra fields	Gauge, spin, relativistic, bath, or QFT load.
No measurement/update selector	Collapse or projection laundering.
Domain discipline	Boundary-condition/self-adjoint-extension laundering.
Potential ledger discipline	Primitive scalar-source import.

Definition 24.2 (First-response Hamiltonian content). *First-response Hamiltonian content is the lowest-order local scalar operator obstruction to spatial standing-preserving continuation inside the*

licensed nonrelativistic projective-Hilbert projection. Because a first spatial derivative either selects a direction, imports a connection/drift, or fails closed scalar self-adjoint standing, the first nontrivial undeclared scalar spatial response is second order.

Definition 24.3 (Minimal same-scope first-response Hamiltonian predicate). *The minimal same-scope first-response Hamiltonian predicate is formally $\text{SameScopeQuantumSpatial}_1(H)$ and is abbreviated $\text{SSQ}_1(H)$. It holds iff H satisfies the following clauses.*

- (Q1) **Hilbert internality:** H is built only from the licensed projective-Hilbert and flat nonrelativistic spatial projection.
- (Q2) **Declared domain:** H acts on a declared dense domain $D(H) \subset \mathcal{H}$.
- (Q3) **Self-adjointness:** H is self-adjoint with $D(H) = D(H^*)$ and $H\psi = H^*\psi$ for all $\psi \in D(H)$, or H is essentially self-adjoint on a declared core and its closure is the self-adjoint generator.
- (Q4) **Linearity:** H is linear on the closed representative state space.
- (Q5) **Projective compatibility:** H respects ray-level state identity and does not make global phase standing-bearing.
- (Q6) **Locality:** H is a local differential operator in the admitted spatial carrier.
- (Q7) **Spatial scalarity:** H is scalar under lawful Euclidean spatial redescription and introduces no preferred direction or anisotropic tensor.
- (Q8) **First-response order:** H contains no spatial derivatives beyond the first nontrivial scalar response, namely second order.
- (Q9) **Uniform principal response:** the second-order principal symbol is proportional to the admitted flat spatial metric and has no variable tensor coefficient unless declared richer scope.
- (Q10) **No hidden gauge connection:** gauge potential terms are absent unless declared as connection load.
- (Q11) **No hidden spin/internal operator:** spin or internal-space matrices are absent unless declared.
- (Q12) **No relativistic mixing:** no time-space symmetric relativistic operator, spinor projection, or second-order time equation is imported.
- (Q13) **No nonlinear state dependence:** no term depends on ψ , $|\psi|^2$, expectation values, or state history unless declared effective/open scope.
- (Q14) **No open-system ledger:** no bath, dissipative, Lindblad, or non-Hermitian sink/source term is hidden inside the closed Hamiltonian.
- (Q15) **No measurement/update selector:** no projection, collapse, record selection, or stochastic update is included as same-scope closed generator content.
- (Q16) **No nonlocal operator:** no fractional Laplacian, integral kernel, inverse operator, or memory term is included.
- (Q17) **No undeclared boundary selector:** boundary conditions and self-adjoint extensions are anchors or external data unless fixed by the declared domain.

(Q18) **Potential ledger discipline:** a zeroth-order scalar term may enter only as declared scalar potential/source ledger V .

(Q19) **Nontrivial spatial-response branch:** the candidate is not merely a homogeneous phase baseline for general spatial dynamics.

Remark 24.4 (Target of the predicate). *The predicate SSQ_1 does not classify all possible quantum theories. It classifies the minimal closed nonrelativistic first-response Hamiltonian class. A theory outside SSQ_1 may be legitimate as richer-scope, effective-scope, open-system, measurement/update, relativistic, gauge, spin, or quantum-field structure, but it is not the theorem target.*

Remark 24.5 (The predicate is not definition by desired conclusion). *The clauses of SSQ_1 are not selected because they yield Schrödinger form. They are the target translation of closed autonomous scalar nonrelativistic projective-Hilbert continuation. Removing Hilbert internality loses the carrier. Removing self-adjoint domain discipline loses closed unitary continuation. Removing Euclidean scalarity admits hidden vector/tensor backgrounds. Removing locality or first-response order admits nonlocal or higher-response scope. Removing the no-update clauses admits measurement or open-system repair. Every clause either preserves the declared target or marks a declared exit from it.*

Predicate clause weakened	Cost to the declared target
Remove self-adjointness	Closed norm/probability continuation is lost; open or effective scope enters.
Remove projective compatibility	Representative-vector privilege or global-phase force appears.
Remove Euclidean scalarity	Preferred direction, anisotropy, curved metric, or background tensor is admitted.
Remove uniform principal response	Variable mass, medium, or curved configuration geometry is admitted.
Permit first-order load	Drift, gauge connection, boost-frame ledger, or vector selector is admitted.
Permit boundary selector	Self-adjoint extension or boundary condition is imported.
Permit nonlinearity	Effective source-state feedback or mean-field scope enters.
Permit nonlocality	The continuation envelope is expanded.
Permit measurement/update term	The closed evolution target is abandoned.
Permit spin, gauge, or relativistic terms	A richer projection scope is entered.

25 Hamiltonian-response exhaustion

The predicate is narrow because the target is narrow. A broader predicate would not refute the theorem; it would define a different theorem object by admitting additional anchors, operator load, open/update ledgers, nonlocal structure, boundary data, or richer projection content.

Before the exhaustion proof begins, fix the comparator boundary. This argument is not canonical quantization, not path integration, and not a Galilei-covariance derivation repackaged. Those comparators operate after a quantum carrier and kinematic assumptions have already been admitted.

The AASC proof first licenses the projective-Hilbert projection from AMetric non-selection, fixes state/phase/norm/operator roles, declares the minimal nonrelativistic spatial envelope, defines the same-scope first-response predicate, and then exhausts the operator candidates.

Definition 25.1 (General local second-order scalar candidate). *Inside the declared scalar envelope $\mathcal{H} = L^2(Q, dx)$, a local linear differential candidate of order at most two has schematic form*

$$H = A^{ij}(x)\partial_i\partial_j + B^i(x)\partial_i + C(x),$$

or, after domain-sensitive integration by parts, an equivalent divergence-form presentation. The coefficients are not automatically admissible. Each coefficient family must be typed as admitted metric/scalar structure, declared tensor/operator load, skin, or hidden selector.

Lemma 25.2 (Linearity and representative operator status). *A same-scope closed Hamiltonian generator must be linear on the representative Hilbert carrier. Nonlinear dependence on ψ adds state-dependent tensor/source load, composition-context dependence, or effective/open scope.*

Proof. The standing-bearing state is a ray, and representative choices are skin modulo the projective quotient. A nonlinear generator depending on a chosen representative, decomposition, normalization convention, density, expectation value, or composite context assigns different continuation behavior to the same projective content unless additional state-dependent load is declared. Under independent composition, such dependence becomes a hidden carrier or source ledger. The closed same-scope branch therefore admits linear representative generators. Nonlinear equations may be physically meaningful as mean-field, effective, stochastic, or open-system dynamics; then they are no longer undeclared same-scope closed generators. $\square$

Lemma 25.3 (Self-adjointness and domain standing). *A closed same-scope Hamiltonian generator must be self-adjoint on its declared domain, or essentially self-adjoint with a declared closure. Otherwise norm/probability standing fails or the system is open/effective.*

Proof. For $i\hbar\partial_t\psi = H\psi$,

$$\frac{d}{dt}\langle\psi, \psi\rangle = \frac{i}{\hbar}\langle H\psi, \psi\rangle - \frac{i}{\hbar}\langle\psi, H\psi\rangle.$$

The expression vanishes on a common domain when H is symmetric there, but closed reversible continuation requires a self-adjoint generator, not merely a formal symmetric expression. If a symmetric operator has multiple self-adjoint extensions, the extension choice is domain/boundary data and must be declared. A hidden extension choice is selector import. Non-self-adjoint generators may represent effective loss/gain, absorbing boundaries, or open dynamics, but not closed same-scope continuation [8, 14]. $\square$

Lemma 25.4 (Principal-symbol scalarity). *If the second-order principal symbol carries no declared tensor load beyond the flat spatial metric δ^{ij} , then*

$$A^{ij}(x) = a\delta^{ij}$$

with constant a .

Proof. The coefficient $A^{ij}(x)$ is a symmetric contravariant tensor field or principal-symbol datum. If it is not proportional to δ^{ij} , it selects anisotropic directions and therefore imports tensor load. If it varies over Q , it supplies a medium, curved configuration metric, variable mass profile, or background scalar/tensor field. If such data are declared, the theory has moved to enriched scope. If they are not declared, they are hidden selectors. The minimal same-scope scalar envelope contains only the flat metric and constant projection coefficient, so the principal symbol is $a\delta^{ij}$ and the second-order slot is $a\Delta$. $\square$

Lemma 25.5 (First-order term classification). *A first-order term*

$$B^i(x)\partial_i$$

is not an undeclared same-scope scalar first-response Hamiltonian term.

Proof. The coefficient B^i is a vector field or connection-like object. If it is standing-bearing, it imports a preferred direction, drift field, gauge connection, moving frame, or boundary/domain datum. If it is pure gauge, the gauge structure must be declared. If it is not paired with the required lower-order correction, it fails symmetry and hence closed standing. If it arises from boundary behavior, the boundary condition is an anchor. Therefore no bare first-derivative term survives inside the minimal scalar envelope. Declared electromagnetic coupling, drift, or connection terms are legitimate enriched-scope structures, not hidden minimal alternatives. $\square$

Lemma 25.6 (Affine momentum and boost ledger classification). *A constant first-order self-adjoint term such as*

$$v \cdot p = -i\hbar v^i \partial_i$$

does not refute the normal form. When paired with the quadratic kinetic branch,

$$\frac{p^2}{2m} + v \cdot p + C = \frac{(p + mv)^2}{2m} + C - \frac{m|v|^2}{2}.$$

It is therefore one of: inertial-frame or Galilean-boost anchor motion; a momentum-origin convention with homogeneous energy-baseline shift; a declared drift/effective ledger; or, if undeclared, a hidden preferred-frame selector. It is not an additional minimal scalar first-response slot.

Proof. The vector v is not supplied by the minimal scalar spatial envelope. If it records a boost or moving frame, the carrier presentation has changed by an inertial-frame ledger; completing the square returns the same quadratic kinetic form plus a baseline energy shift. If v records physical drift, current, or environment motion, it is declared tensor/operator load. If no such declaration is made, the vector selects a preferred direction and momentum origin. Hence the affine momentum term is classified rather than admitted as a new same-scope slot. $\square$

Lemma 25.7 (Zeroth-order scalar slot). *The only same-scope local scalar multiplicative operator with no spatial derivative is*

$$V(x),$$

where V is a declared real scalar potential/source ledger on a domain for which the total Hamiltonian is self-adjoint or has a declared self-adjointness certificate.

Proof. A zeroth-order local operator multiplies the scalar wavefunction by a scalar function. If the multiplier is constant, it contributes a homogeneous energy/phase baseline. If it varies over the admitted spatial carrier, it is scalar potential content. Matrix-valued, stochastic, nonlocal, gauge-coupled, history-dependent, or state-dependent multipliers add structure beyond the minimal scalar envelope. $\square$

Lemma 25.8 (Higher-response and nonlocal exclusion). *Higher derivatives, fractional Laplacians, inverse operators, integral kernels, memory terms, pseudodifferential symbols, and arbitrary functions of Δ are higher-response, effective, or nonlocal continuation-envelope expansions rather than minimal same-scope first-response Hamiltonian content.*

Proof. The predicate admits the first nontrivial scalar spatial response. Operators such as Δ^2 , $f(\Delta)$, $(-\Delta)^\alpha$, inverse operators, integral kernels, time-memory maps, and pseudodifferential symbols require higher spatial response, arbitrary operator-function selection, nonlocal continuation, or effective expansion. These may be legitimate when declared, but they are not undeclared first-response content. $\square$

Lemma 25.9 (Gauge, spin, relativistic, topological, and boundary exclusions). *Gauge connection, spin coupling, relativistic structure, topology-sensitive connection data, and boundary/domain selection are legitimate when declared, but they are not undeclared same-scope minimal first-response Hamiltonian content.*

Proof. Gauge coupling replaces ∂_i by a covariant derivative and imports connection content. Spin coupling introduces internal matrix/operator content. Relativistic wave equations change the spacetime projection, representation class, or time-derivative order. Aharonov–Bohm-type terms import topology/gauge data. Boundary conditions or self-adjoint extension choices are external operator data unless fixed by the declared domain. Each case is richer, effective, declared, or boundary-enriched scope rather than an undeclared same-scope alternative. $\square$

Theorem 25.10 (Hamiltonian-response normal form). *If $\text{SSQ}_1(H)$ in the minimal nonrelativistic scalar spatial envelope, then*

$$H = a\Delta + M_V,$$

where a is a constant projection-envelope coefficient and M_V is the multiplication operator associated with the declared scalar potential/source ledger. The dimensional interpretation of a requires action/inertial scale anchoring.

Proof. By linearity and domain discipline, H is a representative linear operator on a declared dense Hilbert domain. By closed continuation, it is self-adjoint or essentially self-adjoint with declared closure. Locality and first-response order reduce the candidate to the general second-order local form

$$A^{ij}(x)\partial_i\partial_j + B^i(x)\partial_i + C(x).$$

Principal-symbol scalarity gives $A^{ij} = a\delta^{ij}$ with constant a , hence the second-order contribution $a\Delta$. The first-order classification and affine-momentum lemmas exclude $B^i\partial_i$ unless vector, gauge, drift, boost/frame, or domain load is declared, in which case the term is retyped as anchor motion, baseline convention, or enriched/effective scope. The zeroth-order lemma leaves $C = V$ as scalar potential/source ledger. Higher derivative, nonlocal, gauge, spin, relativistic, nonlinear, open, measurement, and boundary-selection alternatives are excluded by the predicate or retyped as declared richer/effective/open/update/boundary scope. Hence every surviving minimal same-scope candidate has normal form $H = a\Delta + M_V$. $\square$

Table 2: Hamiltonian candidate challenge ledger.

Candidate	Disposition inside the minimal first-response target
Constant energy shift	Homogeneous phase/baseline ledger; often skin modulo global phase unless physically anchored.
$V(x)$	Declared scalar potential/source ledger.

Candidate	Disposition inside the minimal first-response target
$A^{ij}(x)\partial_i\partial_j$	Curved metric, medium, variable mass, or tensor principal-symbol load unless $A^{ij} = a\delta^{ij}$.
$B^i(x)\partial_i$	Direction, drift, connection, boost/frame, or boundary-domain import unless declared.
Constant affine momentum $v \cdot p$	Galilean boost/frame ledger, momentum-origin convention, drift/effective load, or hidden preferred-frame selector.
Δ	Same-scope second-order scalar survivor.
Δ^2	Higher-response/effective operator load.
$f(\Delta)$	Arbitrary operator-function selector unless declared.
Fractional Laplacian	Nonlocal/altered continuation envelope.
Magnetic coupling $(p-qA)^2/2m$	Declared gauge connection enrichment.
$\sigma \cdot B$	Spin/internal-role tensor load.
Dirac operator	Relativistic spinor projection.
Nonlinear term $g \psi ^2\psi$	Effective mean-field/source ledger.
Lindblad generator	Open-system ledger.
Projection/collapse	Measurement/update regime.
Boundary-condition variant	Declared domain/self-adjoint-extension anchor.

26 Relation to conventional quantum derivations

This argument is not an invocation of canonical quantization, path-integral formalism, Galilean covariance derivations, information-theoretic reconstructions, or textbook Hilbert-space postulates. Those are useful comparators. They operate after parts of the quantum formalism have already been admitted.

The AASC proof asks prior questions: how state identity obtains standing; why representative-vector identity is quotient/projective; why closed continuation preserves norm/probability; why the generator is self-adjoint; why the minimal spatial generator is exhausted by Δ and V ; and why alternatives must be typed rather than silently admitted. The result is consistent with standard operator-theoretic and Stone-theorem presentations, but those presentations are not used to license the AASC target.

Part VII

Closed Continuation, Scale Anchoring, and the Schrödinger Combination

27 Unitary compatibility and generator form

Theorem 27.1 (Closed continuation generator theorem). *If U_t is a strongly continuous closed reversible standing-preserving one-parameter group on the representative Hilbert projection, then*

$$U_t = e^{-itH/\hbar}$$

for a self-adjoint H .

Proof. Closed trajectory coherence gives transition-probability preservation at ray level. Wigner implementation and continuous-time branch selection give a unitary representative group. Stone's theorem gives a self-adjoint generator. The factor $\hbar$ records the admitted action/phase scale rather than a boundary-generated value. $\square$

Theorem 27.2 (Hamiltonian normal-form insertion). *For the minimal scalar nonrelativistic spatial projection,*

$$\text{SSQ}_1(H) \Rightarrow H = a\Delta + M_V.$$

Proof. This is the Hamiltonian-response normal-form theorem from Part VI. It is repeated here to show the exact insertion point in the closed-continuation chain. $\square$

Theorem 27.3 (Coefficient fixation after inertial/action scale anchoring). *The AASC-native spatial normal-form theorem gives $H = a\Delta + M_V$. The numerical coefficient is fixed only after the nonrelativistic inertial/action scale anchors are admitted. In a nonrelativistic scalar particle projection with admitted action scale $\hbar$, inertial mass anchor m , and free-particle dispersion*

$$E(p) = \frac{|p|^2}{2m},$$

the kinetic coefficient in $H = a\Delta$ is

$$a = -\frac{\hbar^2}{2m}.$$

Proof. Use the Fourier representation rather than treating plane waves as normalizable states. The Laplacian acts spectrally as

$$\Delta \leftrightarrow -|k|^2,$$

and the admitted action/phase scale gives $p = \hbar k$. Thus the kinetic branch $a\Delta$ has spectral value $-a|k|^2$. Matching the admitted nonrelativistic inertial dispersion

$$E = \frac{|p|^2}{2m} = \frac{\hbar^2|k|^2}{2m}$$

requires

$$-a = \frac{\hbar^2}{2m}, \quad a = -\frac{\hbar^2}{2m}.$$

This step is not AMetric numerical generation of m or $\hbar$; it is coefficient fixation after the nonrelativistic inertial/action anchors are admitted. $\square$

Remark 27.4 (Projective Galilei representation witness). *A stronger conventional-facing route identifies mass as the central charge in the projective representation of the Galilei group [16]. The present theorem does not require deriving the Galilei representation from below. It uses the nonrelativistic inertial/transport anchor exactly where coefficient normalization is needed, while preserving the AASC distinction between form standing and numerical scale standing.*

Proposition 27.5 (Degenerate homogeneous branch). *If $a = 0$ in the Hamiltonian normal form, the surviving generator $H = V$ is pure local potential/phase evolution. It may represent a homogeneous baseline or local potential ledger, but it is not the general spatial kinetic response branch. The Schrödinger theorem therefore uses the nontrivial spatial-response branch $a \neq 0$.*

Corollary 27.6 (Closed-source Schrödinger form). *For the closed same-scope nonrelativistic Hamiltonian projection,*

$$i\hbar\partial_t\psi = \left(-\frac{\hbar^2}{2m}\Delta + M_V\right)\psi.$$

28 Potential as source/persistence ledger

The scalar potential is not assumed as primitive environmental substance. It is the Hamiltonian-facing extraction of already admitted source, boundary, or effective roles under the declared scope.

Definition 28.1 (Potential ledger). *A scalar potential V is an admissible source/potential ledger only when it represents a standing-bearing external, environmental, constraint, interaction, or effective role already declared in the closed/effective scope.*

Definition 28.2 (No primitive scalar potential source). *A scalar function $V(x)$ has no source standing merely as syntax. It must be tied to a declared role: external field, constraint, source environment, effective interaction, boundary condition, or calibration ledger.*

Theorem 28.3 (Potential admissibility). *A scalar potential is admissible in the faithful Schrödinger projection iff the associated multiplication operator $M_V : \psi \mapsto V\psi$ is scalar under lawful spatial redescription, multiplicative/local in the minimal scope, source- or boundary-ledger declared, and included in a total Hamiltonian with a declared self-adjoint realization; it is not admissible as a hidden selector, arbitrary fitting function, or undeclared environment import.*

Proof. A potential term changes Hamiltonian standing by a zeroth-order scalar multiplication operator M_V . If M_V is scalar and declared, it is legitimate source/potential ledger content. If it depends on a vector, connection, internal index, stochastic record, bath state, or nonlocal memory, it imports additional operator load. If it is undeclared, it functions as a hidden selector. The total operator $-\hbar^2\Delta/(2m) + M_V$ must have a self-adjoint realization for closed standing. □

Definition 28.4 (Potential/domain self-adjointness certificate). *A scalar potential ledger is domain-certified when the ledger includes conditions under which the multiplication operator $M_V : \psi \mapsto V\psi$ is symmetric on the declared domain and the total operator*

$$H = -\frac{\hbar^2}{2m}\Delta + M_V$$

has a self-adjoint realization, or is essentially self-adjoint after closure on a declared core. A simple sufficient case is real bounded $V \in L^\infty(Q)$ on the standard Laplacian domain; more singular potentials require an explicit certificate, for example relative-boundedness, form-boundedness, Friedrichs selection, or another declared self-adjoint extension route [19, 8].

Theorem 28.5 (Potential ledger self-adjointness condition). *A scalar potential multiplication ledger is admissible in the closed minimal Schrödinger theorem only with a domain/self-adjointness certificate for the total Hamiltonian. Without such a certificate it is not a closed same-scope Hamiltonian term; it is an incomplete source ledger, boundary-domain import, effective approximation, or standing failure.*

Proof. The main theorem requires closed unitary continuation. Closed unitary continuation requires a self-adjoint generator. A formal expression $-\hbar^2\Delta/(2m) + M_V$ does not by itself determine such a generator when V is singular, unbounded in uncontrolled ways, or tied to boundary conditions. Therefore the potential ledger must include the operator-domain certificate that makes the expression a legitimate closed generator. Otherwise the candidate has not supplied the theorem object it claims. □

Boundary/domain choice	AASC status
Full $\mathbb{R}^d$ Laplacian	Minimal spatial anchor.
Periodic box	Declared regulator/domain anchor.
Dirichlet condition	Boundary-domain anchor.
Neumann condition	Boundary-domain anchor.
Robin condition	Boundary parameter anchor.
Self-adjoint extension by deficiency indices	Extension-selection anchor.
Friedrichs extension	Form-order selected extension.
Hidden boundary condition	Inadmissible selector import.

29 Time-dependent potentials and non-autonomous driven ledgers

The main theorem is a closed autonomous normal-form theorem. Its Stone generator step uses a one-parameter unitary group with fixed self-adjoint generator. A time-dependent potential $V(x, t)$ changes the continuation envelope: the evolution is then represented by a propagator family rather than by a single autonomous group $U_t = e^{-itH/\hbar}$.

Proposition 29.1 (Driven-potential classification). *A potential $V(x)$ belongs to the closed autonomous scalar potential ledger when it is time-independent within the declared continuation scope. A term $V(x, t)$ is admissible only as a declared driven, external-source, control, boundary, or effective ledger. It may have an instantaneous Schrödinger-like local form, but it is not a counterexample to the autonomous Stone-generator theorem.*

Proof. Time dependence in the generator supplies information not contained in the closed autonomous state geometry. If the time dependence is physical, it records an external control, source, boundary, or environment ledger. If no ledger is declared, it is hidden continuation import. Thus non-autonomous Schrödinger equations are classified as driven or enriched continuation envelopes, while the present theorem proves the closed autonomous scalar normal form. $\square$

30 Mass/inertia bridge

Definition 30.1 (Particle persistence class). *A particle role is a localized persistence class*

$$\Pi_D = [\kappa_U]_{\equiv \Pi}.$$

Definition 30.2 (Mass and inertial response role). *Mass is the deformation/transport-response profile relevant to nonrelativistic spatial continuation. Inertia is its transport-facing restriction. Neither is primitive scalar substance in the proof target.*

Proposition 30.3 (Mass-to-Hamiltonian bridge). *The kinetic coefficient follows the role chain*

$$\Pi_D \rightarrow \mu_D(\Pi) \rightarrow \iota_D(\Pi) \rightarrow -\frac{\hbar^2}{2m}\Delta.$$

Proof. A particle persistence class supplies the identity target. The mass profile supplies standing-preserving deformation response. The inertial profile supplies transport-state response. The nonrelativistic projective-Hilbert projection represents that transport response by the quadratic

momentum generator $p^2/(2m)$, yielding the Laplacian coefficient after the $p = -i\hbar\nabla$ scale anchor is fixed. $\square$

Corollary 30.4 (No primitive mass coefficient). *The scalar m is Hamiltonian-bearing only as a licensed inertial/transport scale anchor. The theorem derives the form of its occurrence, not its numerical value.*

Part VIII

Phase, Probability, and Scale Ledger

31 Phase baseline

Definition 31.1 (Global phase skin). *Global phase equivalence is*

$$\psi \sim e^{i\theta}\psi.$$

Global phase does not carry standalone predictive standing.

Proposition 31.2 (Global phase is not nullity). *Global phase may change representative notation but does not erase the ray-state content.*

Proof. The ray $[\psi]$ remains the same under multiplication by a nonzero phase. Therefore global phase change is a representative change, not nullification of quantum standing. $\square$

Theorem 31.3 (No primitive phase source). *A phase factor does not carry source or dynamical standing unless relative, gauge-coupled, or otherwise declared as tensor/operator content.*

Proof. A global phase alone changes no projective state, transition probability, or closed Hamiltonian trajectory. If a phase difference affects interference, it is relative phase and is standing-bearing content. If a phase is tied to a gauge connection, the connection is declared operator load. Without such declaration, phase is skin. $\square$

32 Probability/norm quotient discipline

Definition 32.1 (Norm standing). *Closed representative norm encodes probability standing:*

$$\|\psi\|^2 = 1$$

for normalized representatives of rays.

Theorem 32.2 (Homogeneous normalization invariance). *Scaling a nonzero vector representative and then projectivizing preserves ray identity once norm/probability standing is fixed.*

Proof. For any nonzero scalar λ , $\lambda\psi$ determines the same ray. Normalization chooses a representative on the unit sphere after the projective state has already been fixed. It is not a primitive state selector. $\square$

Corollary 32.3 (Probability as quotient-stable predictive ledger). *Probability is not primitive AMetric weight. It is the projective-Hilbert-facing standing ledger of quantum predictive content.*

33 $\hbar$, mass, and numerical scale discipline

Theorem 33.1 (No boundary generation of $\hbar$). *$\hbar$ is a quantum action/phase scale anchor, not an AMetric boundary output in this theorem.*

Proof. The constant $\hbar$ converts between phase/time generator normalization and physical action units. Such conversion requires an admitted comparison scale. The AMetric boundary supplies no unit or calibration ledger. $\square$

Theorem 33.2 (No boundary generation of m). *Mass is an inertial/transport-response anchor, not generated by the Schrödinger-form theorem.*

Proof. The theorem fixes that nonrelativistic kinetic response enters through $p^2/(2m)$. The value of m belongs to particle persistence and inertial transport calibration. It is not produced by the Hamiltonian normal-form exhaustion. $\square$

Quantity	Endpoint status
$\hbar$	Phase/action scale anchor.
m	Inertial/transport scale anchor.
V	Declared scalar potential/source ledger.
Energy zero	Homogeneous phase/baseline convention unless anchored.
Spatial dimension	Projection anchor unless separately derived.
Spin/gauge charge	Declared richer-scope internal/gauge role.

Part IX

Measurement, Decoherence, and Open-System Route Closure

34 Purpose of the measurement/open-system module

Measurement, collapse, decoherence, environment coupling, and stochastic updates are not used to derive the Schrödinger equation. They are route-closure discipline: they cannot be smuggled back as hidden same-scope repairs of closed Schrödinger dynamics.

Definition 34.1 (Measurement/update scope). *A measurement/update scope is a tuple*

$$S_{meas} = (E_{sys}, E_{app}, R_{rec}, W_{obs}, \text{Cont}, Env),$$

where E_{sys} is the evaluated system, E_{app} is the apparatus/environment, R_{rec} is the record relation, W_{obs} is the witness algebra, Cont is the continuation/update rule, and Env is the admissibility envelope.

Definition 34.2 (Certificate-complete measurement scope). *A measurement scope is certificate-complete when every standing-bearing update is represented by an explicit record, environment, apparatus, update rule, or effective ledger.*

Theorem 34.3 (No hidden measurement repair). *A measurement/update response is admissible on the same quantum projection scope only if it is one of: skin-equivalent presentation; declared measurement/update rule; declared environment/open-system ledger; record-bearing boundary certificate; effective stochastic scope; or richer-scope theory. Otherwise it is hidden repair, branch-selection laundering, collapse import, or scope drift.*

Proof. A closed Hamiltonian trajectory has no update selector beyond its generator. If measurement changes standing-bearing state content, it must supply a record, apparatus/environment ledger, stochastic/effective update rule, or richer scope. If it changes no standing, it is skin. If none of these is declared but the state is claimed to jump, the jump is hidden repair or branch selection. $\square$

Corollary 34.4 (No closed-dynamics rescue by collapse). *Collapse cannot repair closed Schrödinger evolution as the same act. It is a new update regime or scope change.*

Corollary 34.5 (Decoherence as declared ledger). *Decoherence is admissible as environment/record/effective ledger, not as hidden same-scope modification of the closed Hamiltonian.*

Part X

Modified Quantum Dynamics Classification

35 Classification theorem

Outside SSQ_1 does not mean false. It means not an undeclared minimal first-response same-scope alternative. Declared richer-scope and effective theories may be legitimate; they are simply not the theorem target.

Theorem 35.1 (Modified-quantum-dynamics classification). *Any proposed modification of Schrödinger form in the faithful projective-Hilbert projection is one of: same-scope collapse to Schrödinger form; skin; anchor change; hidden operator/tensor load; higher-response spatial load; gauge/internal-role enrichment; relativistic projection; nonlocal continuation expansion; open-system ledger; measurement/update regime; boundary-domain/self-adjoint-extension enrichment; richer-scope theory; or standing failure.*

Proof. If the modification satisfies SSQ_1 , the Hamiltonian normal-form theorem forces $H = a\Delta + M_V$ and scale anchoring gives Schrödinger form. If it changes no standing-bearing content, it is skin. If it changes carrier, dimension, domain, connection, boundary, locality, closedness, or representation class, it changes anchors. If it adds fields, matrices, environment records, stochastic maps, nonlocal kernels, or state-dependent terms, it imports operator/tensor load or opens scope. If none of these licensing routes applies, the modification fails standing. $\square$

Table 3: Modified quantum dynamics taxonomy.

Candidate	AASC classification
Free Schrödinger equation	Same-scope survivor.
$V\psi$	Scalar potential/source ledger.

Candidate	AASC classification
$V(x, t)\psi$	Non-autonomous driven, source, control, or effective ledger; not a fixed Stone-generator counterexample.
Constant energy shift	Homogeneous phase/baseline ledger; often skin modulo anchoring.
Minimal electromagnetic coupling	Declared gauge connection.
Pauli equation	Spin/internal-role tensor load.
Dirac equation	Relativistic spinor projection.
Klein-Gordon equation	Relativistic scalar projection.
Nonlinear Schrödinger equation	Effective/source/mean-field scope unless a declared fundamental nonlinear scope is adopted.
Gross-Pitaevskii equation	Effective many-body/mean-field ledger.
Lindblad equation	Open-system continuation ledger.
Collapse model	Measurement/update regime.
Fractional Schrödinger equation	Nonlocal or altered continuation envelope.
PT-symmetric Hamiltonian	Requires altered inner product, open/effective ledger, or declared metric operator.
Boundary-condition variant	Declared domain/self-adjoint-extension anchor.
Quantum-field Hamiltonian	Field-theoretic richer scope.

Part XI

Main Theorem

36 Schrödinger dynamics as the unique minimal first-response Hamiltonian projection

Theorem 36.1 (Main theorem: closed autonomous Schrödinger projection). *Let a nondegenerate determinate admissible quantum predictive interior $\mathfrak{S}$ instantiate the kernel*

$$K = (\text{Adm}, \text{Stand}, \text{Ref}, \text{Irr}),$$

the fixed-domain consequence layer A^+ , and only the exact representative certificate $c_{\mathfrak{S}}^{\text{QP}}$. Import W9.19 first to construct

$$(P_{\mathfrak{S}}, q_{\mathfrak{S}}, \bar{p}_{\mathfrak{S}}, \bar{C}_{\mathfrak{S}}, \bar{\boxtimes}_{\mathfrak{S}}; c_{\mathfrak{S}}^{\text{QS}}).$$

Let a separately constructed faithful closed nonrelativistic projective-Hilbert projection

$$\Pi_{\text{NR}} : P_{\mathfrak{S}} \longrightarrow \mathbb{P}L^2(\mathbb{R}^d, dx)$$

preserve the derived standing, transition weight, continuation, and declared independent-composition law, and use the primary full-space minimal envelope

$$\Pi_H(A_Q) = (\mathbb{R}^d, \delta, dx, \mathcal{H}, \mathbb{P}(\mathcal{H}), \langle \cdot, \cdot \rangle, D(H)), \quad \mathcal{H} = L^2(\mathbb{R}^d, dx).$$

Let closed reversible continuation preserve transition-probability standing on rays, let Wigner implementation, projective phase lifting, and strong continuity give a phase-fixed unitary group U_t , and let Stone's theorem supply its self-adjoint generator H . Let H satisfy $\text{SSQ}_1(H)$ as a nontrivial general spatial-response candidate. Let M_V be the multiplication operator associated with the declared scalar potential/source ledger and with a domain/self-adjointness certificate for the total Hamiltonian, and let $\hbar, m$ be admitted action/inertial scale anchors. Then the unique minimal first-response same-scope closed autonomous Hamiltonian evolution form is

$$i\hbar\partial_t\psi = \left(-\frac{\hbar^2}{2m}\Delta + M_V\right)\psi.$$

Proof. The kernel and A^+ fix target, standing, admissibility, reference, irreversibility, AMetric no-selector discipline, no same-act repair, and no hidden same-scope discriminator. Theorem 8.3 is applied to $c_{\mathfrak{S}}^{\text{QP}}$ and supplies the standing-bearing quotient, separated $\bar{p}_{\mathfrak{S}}$, descended $\bar{C}_{\mathfrak{S}}$ and $\boxtimes_{\mathfrak{S}}$, and their naturality. None is reconstructed later from Hilbert or Hamiltonian data. Those conclusions type the domain and preservation obligations of Π_{NR} ; without W9.19 the displayed projection is not a map from a certified standing object. Faithful projective-Hilbert projection then licenses $(\mathbb{R}^d, \delta, dx, \mathcal{H}, \mathbb{P}(\mathcal{H}), \langle \cdot, \cdot \rangle, D(H))$ as representative geometry and primary full-space spatial envelope, not primitive foundation. Boundary-trace fixation fixes state, phase, norm/probability, operator-domain, potential/source, and measurement/update roles as role-anchor equivalence classes without representative selection. Coherent projected quantum trajectory dynamics gives closed norm/probability and transition-probability continuation. Wigner implementation gives unitary or antiunitary representatives; strong continuous one-parameter time continuation selects the unitary branch, and the projective phase-lift lemma fixes a representative unitary group up to homogeneous energy-baseline ledger. Stone's theorem then gives a self-adjoint generator H and abstract Schrödinger form. The predicate $\text{SSQ}_1(H)$ applies to the spatial Hamiltonian side inside the declared minimal spatial envelope. By Hamiltonian-response exhaustion,

$$H = a\Delta + M_V.$$

The potential ledger includes the domain/self-adjointness certificate required for closed unitary continuation. Nonrelativistic inertial/action scale anchoring fixes

$$a = -\frac{\hbar^2}{2m}.$$

Therefore

$$i\hbar\partial_t\psi = \left(-\frac{\hbar^2}{2m}\Delta + M_V\right)\psi.$$

Any alternative inside the target is skin, anchor motion, hidden operator load, richer scope, effective/open/update scope, boundary-domain enrichment, or standing failure. $\square$

Remark 36.2 (Scope of uniqueness). *Uniqueness is uniqueness of form within the declared closed autonomous scalar nonrelativistic projective-Hilbert projection envelope and up to declared scale anchors $\hbar, m$, potential multiplication ledger M_V , and domain conditions. It is not uniqueness of all quantum dynamics and not exclusion of declared richer-scope theories.*

Corollary 36.3 (Declared-domain extension). *For a declared regular spatial domain $Q \subseteq \mathbb{R}^d$, the same local Hamiltonian form holds only relative to an explicit domain, boundary, and self-adjointness certificate. Dirichlet, Neumann, Robin, periodic, Friedrichs, deficiency-index, and other extension selections are anchors of the declared domain problem, not hidden same-scope alternatives to the primary full-space theorem.*

Proof. Changing from $\mathbb{R}^d$ to $Q \subseteq \mathbb{R}^d$ introduces boundary and extension data. If those data are declared and the total Hamiltonian has a self-adjoint realization, the local scalar first-response form remains $a\Delta + M_V$ on that domain. If they are not declared, the candidate imports a hidden selector and fails same-scope standing. $\square$

Proposition 36.4 (Paired-projection architecture as a unification foothold). *Together with the corresponding Einstein metric-projection theorem [3], the result yields the paired projection schema*

$$\begin{aligned} & \text{AASC standing-preserving continuation} \\ \Rightarrow & \begin{cases} \text{faithful metric projection} \Rightarrow \text{Einstein dynamics,} \\ \text{faithful projective-Hilbert projection} \Rightarrow \text{Schrödinger dynamics.} \end{cases} \end{aligned}$$

This is not yet a full quantum-gravity theory. It is the common normal-form basis for a later unification theorem: the two flagship dynamics are distinct faithful projections of one upstream admissibility/standing/continuation kernel.

Remark 36.5 (Not yet a joint dynamics theorem). *The paired projection schema is deliberately not reported as a full quantum-gravity theory or joint metric/projective-Hilbert dynamics. It is a unification foothold: the two flagship dynamics are no longer independent primitives but distinct faithful projections of the same upstream admissibility/standing/continuation discipline. A later theorem must supply the joint compatibility carrier.*

37 AASC-free reconstruction boundary

A conventional reconstruction is a different theorem with a different proof carrier. It would have to replace the kernel-forced Hamiltonian-projection route by a Hilbert-first, Galilean-first, path-integral, or canonical-quantization theorem proving all of the following under explicit non-AASC assumptions and without hidden imports: fixed state/ray target; lawful redescription; closed norm/probability continuation; self-adjoint generator; spatial first-response exhaustion; potential/source ledger discipline; measurement/open-system route closure; and modified-dynamics classification.

Proposition 37.1 (Reconstruction route is separate). *A Hilbert-first, path-integral, canonical-quantization, or Galilean-covariance derivation is correctness-relevant to this manuscript only if it shows that one of the named AASC links fails or that it proves the same theorem under the same target with fewer declared anchors and without hidden selector, operator, source, continuation, or scope import.*

Proof. A parallel proof route can establish the same conclusion by different means, but it does not invalidate an existing proof unless it exposes an error in the existing proof's assumptions, inferences, or scope report. Since this manuscript declares its target and proof class, an AASC-free reconstruction is correctness-relevant only through a named link failure. $\square$

38 Correctness-relevant denial targets

A same-mode denial is correctness-relevant only if it identifies one of the following failures.

1. Target adequacy does not force the kernel.

2. The AMetric boundary does not block primitive Hilbert/wavefunction backflow.
3. Quantum standing does not require quotient/projective architecture.
4. Global phase is not skin.
5. Projective state identity is not forced.
6. Hilbert projection is improperly licensed.
7. Closed standing-preserving continuation does not force unitarity.
8. Self-adjoint generator form is illicitly assumed.
9. $\text{SSQ}_1(H)$ excludes a genuine same-scope first-response Hamiltonian.
10. Hamiltonian-response exhaustion misses an admissible local scalar operator.
11. The coefficient $-\hbar^2/(2m)$ is illicitly imported rather than scale-anchored.
12. V is misclassified as scalar potential/source ledger.
13. Measurement/open-system terms are same-scope closed-dynamics repairs rather than update/open/effective regimes.
14. Modified-dynamics classification falsely rejects declared richer-scope theories.
15. The theorem overreports beyond closed nonrelativistic projective-Hilbert dynamics.

39 Conclusion

AASC does not begin with a wavefunction, Hilbert vector, Hamiltonian operator, probability amplitude, or Laplacian. It begins with admissibility, standing, reference, irreversibility, lawful redescription, raw predictive comparison, and continuation. This paper imports from W9.19 the standing-bearing quotient, separated transition structure, descended continuation/composition, and naturality absent from its raw c^{QP} premise. It does not rederive those conclusions from later Hilbert machinery. When the resulting standing object is faithfully projected into a closed nonrelativistic projective-Hilbert regime, and when trajectory coherence and boundary-trace role fixation are made explicit, the first-response Hamiltonian candidates are exhausted by

$$H = a\Delta + M_V.$$

Closed state-continuation and scale anchoring fix

$$H = -\frac{\hbar^2}{2m}\Delta + V.$$

Therefore

$$i\hbar\partial_t\psi = \left(-\frac{\hbar^2}{2m}\Delta + M_V\right)\psi$$

is the unique minimal first-response same-scope Hamiltonian projection of closed AASC quantum standing.

The result mirrors the Einstein dynamics theorem at the proof-class level. Einstein dynamics is the unique minimal first-response metric projection of AASC gravity; Schrödinger dynamics is the unique minimal first-response Hamiltonian projection of AASC quantum standing. The paired result is the natural foundation for a later unification manuscript.

Appendices

A Provenance support map

Table 4: Provenance support map.

Support cluster	Function in the Schrödinger projection theorem
Quantum physical quotient structure	Full-source expansion of the quantum predictive regime, Hilbert-or-denial theorem, inner-product reconstruction, unitary/self-adjoint generator form, and quantum interior rigidity.
WDW-9, Theorem W9.19	Load-bearing construction of standing descent, separated quotient comparison, descended continuation/composition, and naturality from the exact raw certificate.
WDW-10, Theorem W10.25	Occupied positive clock-neutral rank-one realization witness; no spatial-carrier transfer.
Realized physical interior	Bridge showing Hilbert/Schrödinger quantum interior as part of the realized physics stack.
Same-scope operator exhaustion	Operator-totalization, hidden import, repair, and same-scope closure discipline.
Self-adjoint extension selection	Operator-domain, boundary-condition, Friedrichs, and fixed-base extension discipline.
Boundary-trace fixation	Role-anchor fixation without representative selection for state, phase, norm, operator, potential, and measurement roles.
Coherent trajectory theory	Legal trajectory, coherent trajectory, failure witness, no deferred coherence, and no skin repair.
Present paper	$SSQ_1(H)$, Hamiltonian spatial-response exhaustion, scale/potential ledger, measurement/open route closure, and modified-dynamics taxonomy.

B Formal predicate $SSQ_1(H)$

The predicate $SSQ_1(H)$ defines the minimal closed nonrelativistic first-response Hamiltonian target. It is intentionally narrow. It does not prohibit higher-derivative, metric-affine, spin, gauge, relativistic, nonlinear, nonlocal, stochastic, or open-system theories; it classifies them outside the minimal closed first-response target unless they collapse to the normal form.

- (S1) Hilbert-internality: H is built only from the licensed scalar projective-Hilbert projection and declared spatial carrier.
- (S2) Declared domain: H is defined on a dense domain $D(H)$ with closure/extension status fixed.
- (S3) Self-adjointness: H is self-adjoint or essentially self-adjoint with declared closure.
- (S4) Linearity: H is linear on the representative Hilbert carrier.
- (S5) Projective compatibility: global phase remains skin and ray identity is preserved.
- (S6) Locality: H is a local differential operator.
- (S7) Spatial scalarity: no preferred vector, tensor, frame, or anisotropic principal symbol appears.
- (S8) First-response order: no derivative order beyond the first nontrivial scalar spatial response.
- (S9) Uniform principal response: the principal symbol is proportional to the admitted flat metric.
- (S10) No hidden gauge, spin, relativistic, nonlinear, open, measurement, nonlocal, or boundary-selection load.
- (S11) Potential ledger: the only zeroth-order nonhomogeneous survivor is a declared scalar potential V .

C Hamiltonian-response exhaustion proof

Start with a local scalar linear differential operator of order at most two on the declared scalar spatial carrier:

$$H = A^{ij}(x)\partial_i\partial_j + B^i(x)\partial_i + C(x).$$

Spatial scalarity and no hidden anisotropic tensor force $A^{ij}(x) = ah^{ij}$ with constant a . The first-derivative coefficient B^i is a vector field. If it is nonzero and standing-bearing, it imports a direction, drift, gauge connection, or boundary-domain datum; if it is pure gauge, the gauge ledger is declared; if it fails the divergence correction needed for symmetry, it violates self-adjointness. Thus no undeclared $B^i\partial_i$ term survives. The scalar $C(x)$ is the potential/source ledger $V(x)$, with constants treated as homogeneous phase/baseline terms. Therefore

$$H = a\Delta + M_V.$$

All higher derivative, fractional, nonlocal, spin, gauge, relativistic, nonlinear, open, and update terms lie outside the predicate unless declared as richer/effective scope.

D Unitary generator and coefficient discipline

Closed reversible standing-preserving continuation gives a strongly continuous one-parameter unitary group U_t . Stone's theorem gives $U_t = e^{-itH/\hbar}$ and hence $i\hbar\partial_t\psi = H\psi$. The Hamiltonian-response theorem gives $H = a\Delta + M_V$. The nonrelativistic kinetic response of a particle persistence class is $p^2/(2m)$, with $p = -i\hbar\nabla$. Therefore

$$\frac{p^2}{2m} = -\frac{\hbar^2}{2m}\Delta,$$

fixing $a = -\hbar^2/(2m)$.

E Boundary-trace and coherent quantum trajectory details

Boundary-trace fixation supplies role-anchor equivalence classes without selecting representatives:

$$[\text{StateRole}]_{\equiv \text{Role}}, \quad [\text{PhaseRole}]_{\equiv \text{Role}}, \quad [\text{NormLedgeRole}]_{\equiv \text{Role}},$$

$$[\text{OperatorDomainRole}]_{\equiv \text{Role}}, \quad [\text{PotentialLedgeRole}]_{\equiv \text{Role}}, \quad [\text{MeasurementUpdateRole}]_{\equiv \text{Role}}.$$

This prevents the state, phase, probability, domain, potential, and update roles from being primitive representatives or hidden selectors. A coherent quantum trajectory is a legal same-scope trajectory that preserves standing throughout its retained lineage. No deferred coherence, no skin repair, and irrecoverability of failure block post-hoc source, measurement, or normalization repair.

F Potential/source-ledger extraction

A scalar potential V is the Hamiltonian-facing representation of declared source, interaction, constraint, boundary, or effective environment roles. In the minimal closed target it enters only as a local multiplicative scalar operator. If a proposed V depends on hidden records, state amplitudes, nonlocal memory, gauge connection, spin, relativistic correction, or unregistered environment, it is no longer the minimal scalar potential ledger.

G Phase/probability quotient ledger

Global phase is skin:

$$\psi \sim e^{i\theta} \psi.$$

Relative phase is standing-bearing when it changes interference or transition weights. Norm is the representative ledger of projective probability standing. Normalization chooses a representative of a ray after the ray has standing; it is not primitive state creation.

H Entanglement and boundary compatibility

Entanglement illustrates boundary compatibility rather than hidden same-scope Hamiltonian transport. Joint tensor compatibility can be fixed at admission before spatial communication is relevant; later metric or Hilbert descriptions may show nonclassical correlations, but no hidden Hamiltonian term follows unless a declared interaction, record, gauge, environment, or nonlocal ledger is introduced.

I Measurement/open-system route closure

A measurement or open-system response is admissible only through a declared route: apparatus/environment ledger, record certificate, stochastic update rule, non-Hermitian effective generator, Lindblad open-system generator, or richer scope. Collapse, decoherence, and record selection cannot repair a closed Hamiltonian trajectory as the same act.

J Modified quantum dynamics taxonomy

Table 5: Extended modified quantum taxonomy.

Candidate	AASC classification
$H = -\hbar^2 \Delta / (2m)$	Same-scope free survivor.
$H = -\hbar^2 \Delta / (2m) + V$	Same-scope survivor with declared scalar potential ledger.
$H = a\Delta + M_V$	Pre-scale normal form.
$H = V$ only	Homogeneous/zeroth-order branch; not general spatial-response dynamics.
First-order drift	Preferred vector/drift/gauge/domain load.
Anisotropic kinetic tensor	Declared metric/material tensor or scope change.
Variable mass	Effective/material/source ledger or altered kinetic anchor.
Gauge covariant derivative	Declared connection load.
Spin coupling	Internal-role operator load.
Relativistic Hamiltonian	Relativistic projection.
Quantum field Hamiltonian	Field-carrier richer scope.
Nonlinear dynamics	Effective/source/richer scope.
Stochastic dynamics	Update/effective/open scope.
Non-Hermitian Hamiltonian	Open/effective ledger unless a new inner product/domain is declared.
Boundary extension choice	Declared domain/self-adjoint extension data.

K Scale and endpoint discipline

AASC does not treat dimensional numerical values as primitive outputs. A dimensional value has standing only relative to a scale anchor, comparison carrier, calibration ledger, or source ledger.

Quantity	Endpoint status
$\hbar$	Action/phase scale anchor; numerical value requires calibration.
m	Inertial/transport scale anchor; numerical values are matter-sector data.
V	Potential/source ledger; magnitude requires source calibration.
Energy zero	Homogeneous phase/baseline convention unless fixed by comparison.
Spatial ric/dimension	met- Projection anchors unless separately derived.
Domain	Operator anchor; boundary data must be declared.

L Standalone foundation theorem ladder

Table 6: Foundation theorem ladder.

Foundation link	Schrödinger-theorem function
Target adequacy $\Rightarrow K$	Closed quantum-continuation theorem act already requires admissibility, standing, reference, and irreversibility.
$K \Rightarrow A^+$	Supplies AMetric no-selector boundary, no same-act repair, no second same-scope classifier, report preservation, and no hidden carrier transfer.
Raw c^{QP} certificate $\xrightarrow{\text{W9.19}} c^{\text{QS}}$	Constructs the standing quotient and the descended/natural transition, continuation, and composition structures used by Π_{NR} .
Projective-state corollary	Fixes ray state standing and global phase skin from the imported boundary.
Faithful Hilbert projection	Separately represents predictive geometry without Hilbert backflow or WDW carrier transfer.
Closed continuation	Forces unitary/self-adjoint generator.
SSQ ₁ exhaustion	Performs the new Hamiltonian normal-form proof.
Scale/potential ledger	Fixes $-\hbar^2/(2m)$ and V without boundary-generated constants.

M Adversarial audit matrix

Table 7: Adversarial audit matrix.

Objection	Standing status	Response and named link
AASC is optional vocabulary.	No, unless a role failure is shown.	Closed quantum continuation already fixes target, standing, reference, admissibility, irreversibility, continuation, and reportability. Link: kernel.
W9.19 is optional provenance.	Standing if its conclusions are already premises or independently reproved.	Theorem 8.3 is the sole source for the standing quotient and four descended/naturality structures that type Π_{NR} . Deleting it leaves that projection without a certified domain and preservation laws.
The WDW and Schrödinger signatures differ.	Standing if any raw certificate or derived conclusion differs.	The projects include the same byte-identical <code>qs_boundary.tex</code> ; all separate Hilbert, Wigner–Stone, domain, and SSQ ₁ burdens occur downstream.
The rank-one WDW line is enlarged to $L^2(\mathbb{R}^d)$.	Standing if a carrier map is used.	W10.25 is only an occupied realization witness; the faithful spatial carrier is independently constructed and no WDW-to- L^2 map is claimed.

Objection	Standing status	Response and named link
The downstream paper proves the WDW endpoint.	Standing if dependency polarity reverses.	W9.19 and W10.25 precede this theorem; all spatial projection and Hamiltonian burdens remain downstream and are not used in W9.1–W10.25.
The theorem assumes Hilbert space.	Standing if Hilbert projection is used primitively.	Hilbert structure is licensed as representative projection after quantum standing; the AASC provenance record supplies the Hilbert-or-denial route. Link: Hilbert projection.
The theorem assumes unitarity.	Standing if closed trajectory proof is absent.	Unitarity follows from closed reversible standing-preserving continuation. Link: closed continuation.
The theorem assumes self-adjoint H .	Standing if imported without closure.	Self-adjointness is generator compatibility for closed unitary continuation and domain discipline. Link: self-adjoint generator.
The theorem assumes the Laplacian.	Standing if no exhaustion proof is provided.	The new theorem exhausts local scalar first-response operators. Link: Hamiltonian exhaustion.
Nonlinear quantum mechanics refutes it.	Standing only if same-scope closed minimal.	Nonlinear terms are effective/source/richer scope unless they collapse to linear form. Link: modified taxonomy.
Measurement collapse refutes it.	No for the closed target.	Collapse is update/measurement regime, not closed continuation. Link: measurement route closure.
Lindblad dynamics refutes it.	No for the closed target.	Lindblad dynamics is an open-system ledger. Link: open-system caveat.
Dirac equation refutes it.	No.	Dirac is relativistic spinor projection. Link: richer scope classification.
Gauge coupling refutes it.	No.	Gauge is declared connection load. Link: operator-load classification.
Constants are not predicted.	Not an objection to this target.	$\hbar$ and m require scale anchors; form standing is the theorem target. Link: scale endpoint discipline.

N ATS/UEAP audit and objections

Object	ATS role	
AMetric boundary	Anchor.	
Projective-Hilbert projection	Anchor.	
Closed nonrelativistic scope	Anchor.	
Operator domain	Anchor.	
$[\psi]$	Projective state content.	
H	Operator content.	
Δ	Spatial first-response operator.	
V	Potential/source ledger.	
Global phase	Skin.	
Basis choice	Skin.	
Measurement record	Declared update/record ledger.	

Requested report	Carrier required	Status in this paper
Minimal Schrödinger form	Quantum standing, Hilbert projection, closed continuation, Hamiltonian exhaustion.	Certified by main theorem.
All quantum mechanics	Measurement, QFT, relativistic, and open-system completions.	Not reported.
Numerical value of $\hbar$	Scale calibration.	Out of target.
Numerical mass values	Matter-sector scale anchors.	Out of target.
Measurement solution	Record/update theory.	Not reported.
QFT completion	Field-theoretic projection.	Not reported.
Quantum-gravity unification	Joint metric/Hilbert projection theorem.	Prepared, not claimed.

O Paired-projection unification readiness

The present theorem is intentionally parallel to the Einstein metric-projection theorem. The two normal-form results share the same upstream governance structure but differ in their faithful projection targets.

Einstein dynamics theorem	Schrödinger dynamics theorem
AASC gravity role	AASC quantum standing role
Faithful metric projection	Faithful projective-Hilbert projection
SameScopeMetric ₁ ($E_{\mu\nu}$)	SSQ ₁ (H)
Curvature-response exhaustion	Hamiltonian-response exhaustion
$E_{\mu\nu} = aR_{\mu\nu} + bRg_{\mu\nu} + cg_{\mu\nu}$	$H = a\Delta + M_V$
Divergence/source compatibility	Unitary/norm-standing compatibility
Einstein equation	Schrödinger equation
Modified-gravity taxonomy	Modified-quantum-dynamics taxonomy

The unification opportunity is therefore not an equation-level bolt-on. It is a projection-level theorem program: show how the metric and projective-Hilbert projections share a common admissibility/standing/continuation origin, then characterize any joint richer-scope object that preserves both without hidden backflow.

P Wigner–Stone bridge audit details

The Wigner–Stone bridge is included to prevent a common false compression: norm preservation alone does not license linear unitary dynamics. The standing-bearing quantum object is projective transition structure. A ray transformation preserving transition probabilities is implemented by a unitary or antiunitary representative. Continuous time evolution through the identity selects the unitary branch; Stone’s theorem then supplies the self-adjoint generator. The proof roles are therefore separated as follows.

Bridge step	Proof role
Ray/projective state identity	Removes primitive vector representative standing.
Transition-probability preservation	Supplies the invariant object for Wigner implementation.
Wigner theorem	Gives unitary or antiunitary representative implementation.
Strong one-parameter continuity	Excludes antiunitary continuous time branch.
Stone theorem	Gives self-adjoint generator H .
AASC role discipline	Prevents measurement/open/boundary repair from being hidden in the closed branch.

Q Spatial-envelope proof audit

The normal form $H = a\Delta + M_V$ is deliberately scope-bound. It assumes the minimal scalar nonrelativistic spatial envelope, not an arbitrary configuration geometry. The following failure modes are therefore not counterexamples.

Table 8: Spatial-envelope audit table.

Alternative	Why it is not an undeclared same-scope survivor
Laplace–Beltrami operator	Requires a curved configuration metric anchor.
Variable mass tensor	Requires medium/inertia tensor load.
Magnetic Schrödinger operator	Requires gauge connection.
Aharonov–Bohm term	Requires topology and gauge holonomy data.
Pauli Hamiltonian	Requires internal spin carrier.
Dirac Hamiltonian	Requires relativistic spinor projection and matrix operator structure.
Lindblad generator	Requires open-system ledger.
Nonlinear mean-field Hamiltonian	Requires effective many-body/source-state ledger.
Boundary-dependent Hamiltonian	Requires declared domain/self-adjoint extension.
Fractional Laplacian	Requires nonlocal or altered spatial continuation envelope.

R Programmatic joint-projection target

The later unification theorem should not try to identify the Einstein and Schrödinger equations by syntactic fusion. The correct next target is a joint projection theorem: identify the common admissibility/standing/continuation carrier, prove that metric and projective-Hilbert projections are both faithful, and then classify interaction, measurement, semiclassical, or quantum-field structures as declared cross-ledgers rather than hidden backflow. The present paper supplies one side of that paired architecture; the Einstein dynamics manuscript supplies the other.

S Full proof-link ledger

This ledger records the named proof links that a correctness-relevant objection must attack. It is intentionally explicit because the manuscript is not evaluated by resemblance to a Hilbert-space-first derivation. A conventional reconstruction can be helpful, but it becomes a refutation only by breaking a link below or by proving the same target with fewer declared anchors and no hidden imports.

Table 9: Proof-link ledger for hostile audit.

Proof link	Carrier	Failure blocked
Target adequacy	Determinate closed quantum-continuation act	Treating equations as syntax without standing.
Kernel activation	$K = (\text{Adm}, \text{Stand}, \text{Ref}, \text{Irr})$	Loss of target, status, admissibility, or act-time finality.
AMetric no-selection	Boundary discipline	Primitive Hilbert/vector/operator backflow.

Proof link	Carrier	Failure blocked
Quantum quotient architecture	Projective/ray standing	Representative-vector privilege.
W9.19 structural import	Five quotient-level conclusions from c^{QP}	Optional provenance, a conclusion smuggled into the premise, or an independently duplicated descent proof.
W10.25 realization witness	Clock-neutral rank-one WDW export	Carrier enlargement or false identification with the spatial target.
Transition-probability standing	Ray comparison ledger	Norm-only shortcut to unitary dynamics.
Quantum-interior reconstruction ladder	Restated theorem-use ladder	Treating Hilbert geometry as primitive.
Faithful projective-Hilbert projection	Role-preserving projection	Hilbert facts authorizing their own boundary.
Spatial envelope lock	$\mathcal{E}_{NR}$	Undefined configuration-space counterexamples.
Boundary-trace fixation	Role-anchor equivalence	Representative, basis, domain, or phase selection.
Coherent trajectory theory	Standing-preserving legal trajectories	Deferred normalization or collapse repair.
Wigner implementation	Transition-preserving ray symmetry	Nonlinear norm-preserving map gap.
Continuous branch selection	One-parameter time flow	Antiunitary time-continuation branch.
Stone generator	Strongly continuous unitary group	Self-adjoint generator import without proof role.
$\text{SSQ}_1(H)$ predicate	Fixed target constraints	Hidden operator load or scope drift.
Principal-symbol scalarity	Flat spatial carrier	Variable tensor, medium, or anisotropic import.
First-order exclusion	Vector/gauge/drift classification	Hidden connection or preferred direction.
Zeroth-order slot	Potential/source ledger	Arbitrary scalar selector import.
Domain certificate	Self-adjointness discipline	Boundary/self-adjoint-extension laundering.
Coefficient anchoring	$\hbar, m, E = p^2/(2m)$	Pretending numerical scale is AMetric output.
Modified-dynamics taxonomy	Scope classification	False refutation by richer/open/update theories.

T Kernel-denial cost matrix for the Schrödinger theorem

Kernel denial does not preserve the same Schrödinger theorem with fewer assumptions. It changes or destroys the determinate closed quantum-continuation object.

Table 10: Cost of kernel denial in the Schrödinger theorem.

Denied role	Schrödinger-specific cost
Reference	No fixed state/ray target; no stable meaning for ψ , $[\psi]$, $\mathcal{H}$, $D(H)$, H , V , Δ , spatial carrier, transition probability, or continuation.
Standing	The equation remains formal syntax, modeling convention, or comparator presentation but lacks theorem-bearing closed-continuation force and licensed downstream reuse.
Admissibility	No boundary separates closed evolution from measurement update, collapse, open-system dynamics, nonlinearity, gauge import, spin import, domain selection, boundary condition, stochastic record, or hidden Hamiltonian selector.
Irreversibility	Later normalization, projection, collapse, decoherence, measurement record, domain repair, or branch selection can be laundered into same-act standing.

Corollary T.1 (No kernel-free same theorem). *There is no position from which one may reject the kernel while still using Schrödinger dynamics as a determinate same-scope closed quantum-continuation theorem object. Either the target remains nondegenerate, in which case the kernel roles are active, or one of the costs in the table is paid and the same theorem object is lost.*

U Detailed spatial normal-form case analysis

The central exhaustion step can be read as a classification of the coefficients in the general local candidate

$$H = A^{ij}(x)\partial_i\partial_j + B^i(x)\partial_i + C(x).$$

The table below records the full case split. It is not a rejection of the mathematical legitimacy of the rejected terms. It is a same-scope typing result.

Table 11: Case analysis for the spatial normal form.

Candidate datum	If declared	If undeclared
$A^{ij} = a\delta^{ij}$	Minimal kinetic principal symbol.	Survives as $a\Delta$.
A^{ij} anisotropic	Medium/tensor inertia load.	Hidden direction selector.
$A^{ij}(x)$ variable	Curved/medium/variable-mass envelope.	Hidden background tensor.
B^i vector field	Drift/vector-load theory.	Preferred-direction import.
B^i from gauge potential	Gauge connection enrichment.	Hidden connection.

Candidate datum	If declared	If undeclared
Divergence correction	Domain/symmetry certificate.	Formal self-adjointness gap.
$C(x)$ real scalar	Potential/source ledger V .	Arbitrary fitting function.
C constant	Energy-baseline ledger.	Global phase skin absent energy anchor.
Matrix-valued C	Internal/spin/multiplet load.	Hidden internal carrier.
History-dependent C	Open/effective memory.	Hidden nonlocal continuation.
State-dependent $C(\psi)$	Mean-field/effective source.	Nonlinear same-scope import.
Boundary condition	Domain anchor.	Selector laundering.
Self-adjoint extension	Extension anchor/certificate.	Incomplete theorem object.

V Full modified quantum-dynamics taxonomy

Table 12: Expanded modified-dynamics taxonomy.

Candidate or family	AASC classification
Free scalar Schrödinger equation	Same-scope survivor.
Scalar potential term $V\psi$	Declared scalar potential/source ledger.
Constant energy shift	Homogeneous phase/baseline ledger; standing-bearing only with energy-reference anchor.
Electromagnetic minimal coupling	Declared gauge connection and scalar gauge/source ledger.
Aharonov–Bohm Hamiltonian	Topology/gauge holonomy enrichment.
Pauli equation	Spin/internal-role operator load.
Spin-orbit coupling	Relativistic/effective internal-role correction.
Dirac equation	Relativistic spinor projection.
Klein–Gordon equation	Relativistic scalar projection, different time-order target.
Gross–Pitaevskii equation	Effective many-body/mean-field source ledger.
Nonlinear Schrödinger equation	Nonlinear source/effective/state-feedback scope.
Hartree equation	Effective interaction/source ledger.
Lindblad equation	Open-system continuation ledger.
Non-Hermitian effective Hamiltonian	Loss/gain/open/effective ledger.
PT-symmetric Hamiltonian	Altered metric/operator ledger or open/effective representation.
GRW/CSL collapse	Stochastic measurement/update richer scope.
Projection postulate	Measurement/update regime with record/certificate burden.
Decoherence model	Environment/record ledger.
Fractional Schrödinger equation	Nonlocal/altered spatial continuation envelope.
Pseudodifferential Hamiltonian	Enriched symbol or nonlocal scope.

Candidate or family	AASC classification
Hamiltonian on curved configuration space	Curved configuration metric anchor.
Variable-mass Hamiltonian	Medium/inertial tensor or scalar load.
Boundary-condition Hamiltonian	Domain/self-adjoint-extension anchor.
QFT Hamiltonian	Field-theoretic richer scope.
Semiclassical gravity coupling	Mixed metric/projective-Hilbert ledger; requires cross-carrier certificate.
Path integral presentation	Comparator/skin route unless used to import new anchors.
Canonical quantization route	Reconstruction route, not proof authority for this target.
Pilot-wave guiding equation	Additional ontology/trajectory ledger beyond closed wave-function dynamics.
Many-worlds branching description	Interpretive/reporting layer unless it changes dynamics.
Relational or QBist update	State-ascription/update/reporting regime, not closed generator.

W AASC-free reconstruction checklist

A conventional reconstruction can be valuable. It becomes correctness-relevant only if it proves the same target while preserving the same fixed comparison class or if it breaks a named proof link. The following checklist records what such a reconstruction must supply without hidden imports.

1. A fixed state/ray target and a stable representative quotient.
2. Transition-probability standing rather than primitive vector labels.
3. A lawful-redescription account for basis changes and global phase.
4. A closed/open/update distinction that blocks collapse or environment repair inside closed evolution.
5. A Wigner-type implementation step or equivalent projective symmetry bridge.
6. A Stone-type self-adjoint generator result with domain assumptions.
7. A minimal spatial envelope or an explicit alternative carrier.
8. An operator normal-form theorem for the spatial generator.
9. A domain/self-adjointness certificate for the potential and boundary conditions.
10. A scale/inertial coefficient rule that does not claim boundary generation of $\hbar$ or m .
11. A taxonomy explaining spin, gauge, relativistic, nonlinear, open, nonlocal, and measurement variants.

X Expanded adversarial objections with standing filters

Table 13: Expanded hostile-referee standing filter.

Objection	Standing status	Patch response
Norm preservation does not imply unitarity.	Standing unless patched.	Wigner–Stone bridge now separates transition-preserving ray symmetry from unitary implementation.
Deleting W9.19 leaves the theorem unchanged.	Standing unless the import supplies unavailable content.	The main theorem’s Π_{NR} has domain $P_{\mathfrak{S}}$ and must preserve $\bar{p}_{\mathfrak{S}}$, $\bar{C}_{\mathfrak{S}}$, and $\bar{\mathfrak{Q}}_{\mathfrak{S}}$; all are W9.19 conclusions absent from c^{QP} .
The shared signature only matches by name.	Standing unless theorem boundaries are exact.	The three rebuilt projects use the same byte-identical <code>qs_boundary.tex</code> , separating the six raw ζ -proofs from five derived δ -conclusions.
The WDW realization supplies the spatial carrier.	Standing if a carrier map occurs.	The reciprocal non-export ledger forbids carrier transfer; $L^2(\mathbb{R}^d)$ and all spatial structure are separately constructed downstream.
The theorem assumes Hilbert space.	Standing only if primitive.	Hilbert geometry is restated in the quantum-interior theorem ladder and supported by prior AASC provenance results.
The theorem assumes self-adjointness.	Standing if no generator theorem.	Stone theorem and domain certificate now carry the role.
The Laplacian is assumed.	Standing if no normal form.	Principal-symbol scalarity and first-order exclusion prove the spatial normal form.
The coefficient imports classical kinetic energy.	Standing if reported as boundary output.	Coefficient is now explicitly fixed by admitted nonrelativistic inertial/action anchors.
Plane waves are not L^2 .	Standing if used naively.	Fourier spectral representation replaces normalizable plane-wave reliance.
Singular potentials break self-adjointness.	Standing if no domain rule.	Potential/domain certificate is added.
Gauge coupling refutes it.	No in minimal scope.	Gauge is declared connection enrichment.
Spin refutes it.	No in scalar scope.	Spin is internal-role load.
Relativistic equations refute it.	No.	They change projection class.
Lindblad dynamics refutes it.	No for closed target.	It is open-system ledger.
Collapse refutes it.	No for closed target.	It is update/measurement regime.
Nonlinear equations refute it.	Standing only if same-scope minimal.	Classified as effective/source/state-feedback unless declared otherwise.
Boundary conditions refute it.	No if declared.	They are domain anchors.

Objection	Standing status	Patch response
Unification claim is over-reported.	Standing if called full theorem.	Downgraded to paired-projection foothold.

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